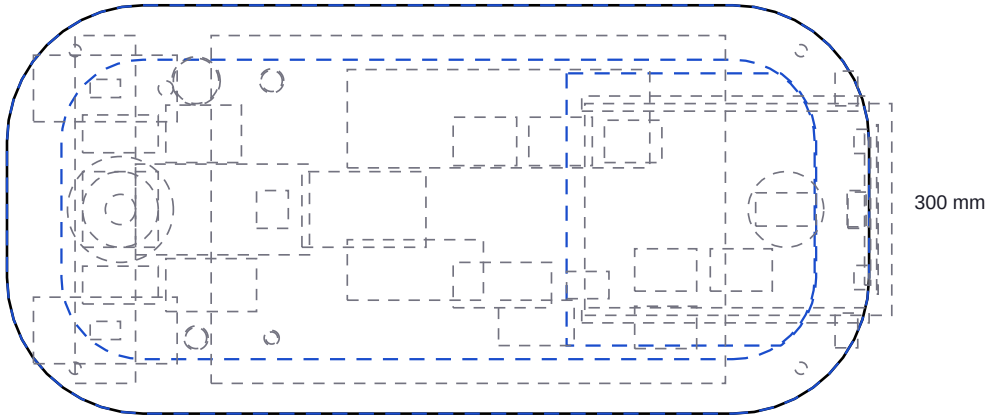
<- must measure 50 mm

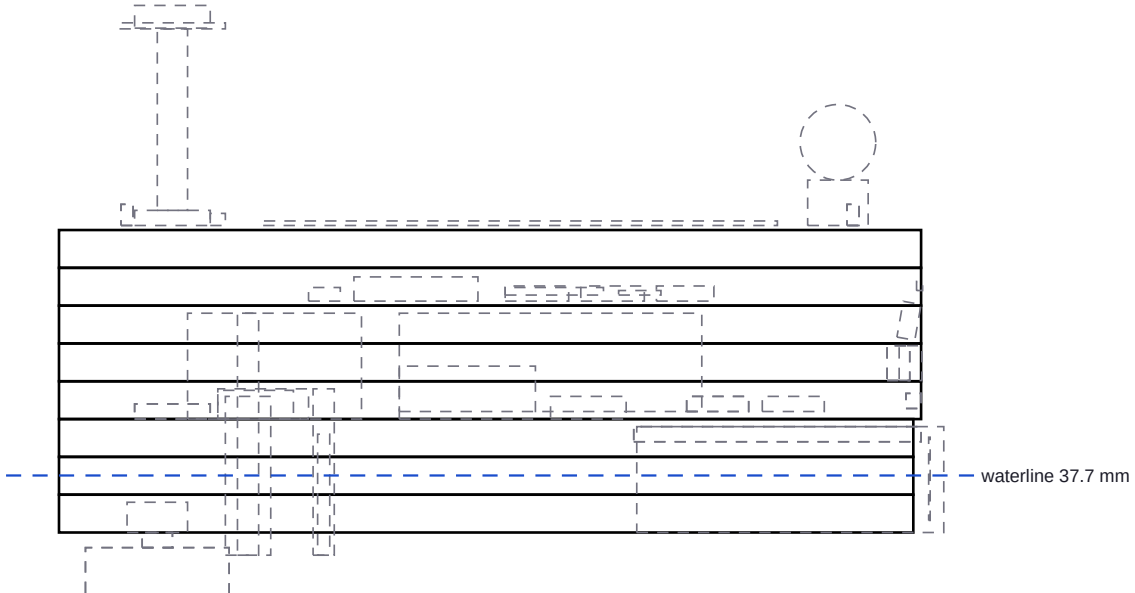
VARUNA AI Pod

600 x 300 mm thermocol boat | Your design, upgraded

PLAN (from above), scale 1:5



SIDE, scale 1:5



Size	600 x 300 mm, hull 200 mm tall, 348 mm to GPS top
Weight	4184 g empty, 4565 g with 381 g of plastic
Draft	37.7 mm empty, 40.9 mm full
Reserve buoyancy	10.5 kg before the deck goes under
Stability (GM)	104 mm: very stable
Trim	bow 2.6 mm higher than stern empty; slide the battery to level
Collection	1.9 L pull-out drawer
Camera	102 mm above water, tilted 10 deg down
Battery	Li-ion 3S6P 20 Ah (222 Wh), about 4.4 h
Thermocol	8 sheets 1000 x 500, or 3 sheets 2 x 3 ft (610 x 915)

How to use these templates

1. Print at 100% / Actual size. Check the 50 mm bar with a ruler.
2. Join tiles by matching the grey 50 mm grid and the tile letters.
3. Stick the paper on thermocol with a glue stick.
4. Cut with a hot-wire cutter or a new snap-off blade, held upright.
5. Stack layers on skewers through the + holes. Glue with Fevicol SH.
6. Never use Fevikwik, spray paint or solvent glue: they melt thermocol.
7. Sand the bow steps smooth, coat with PVA + tissue or packing tape, then water-based acrylic paint.
8. Float-test in a tub before fitting the electronics.

cut line

cut out (window / pocket)

fold / glue line

placement guide, do not cut

skewer hole: lines up the layers

<- must measure 50 mm



ruler in mm: the 0 to 150 marks must be exactly 150 mm apart

100 mm x 100 mm

How to print on A4 so every part comes out real size

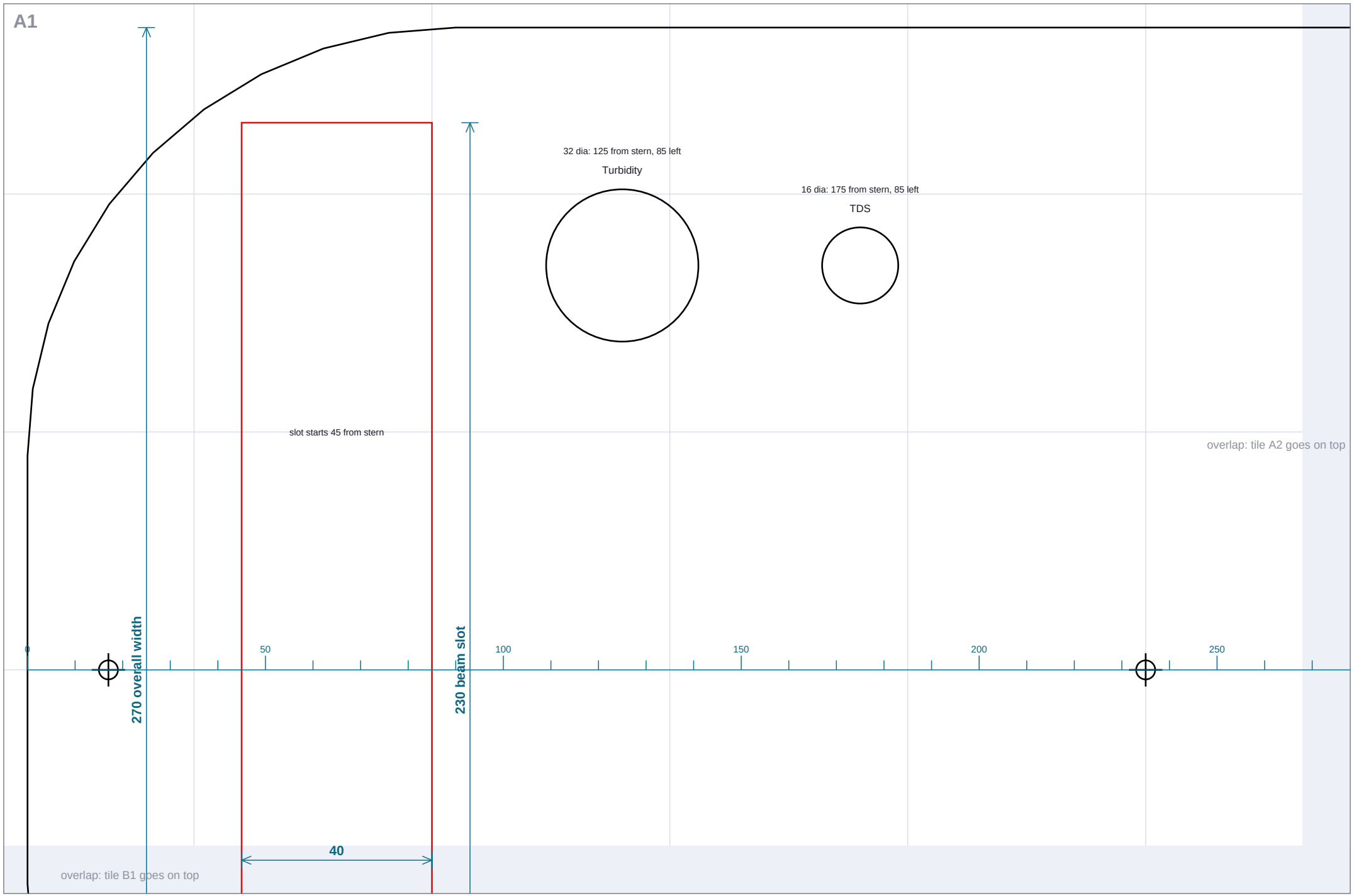
1. Paper: A4, landscape. Printer: 'A4' selected as the paper size.
2. Scaling: 'Actual size' or '100%'. Turn OFF 'Fit to page', 'Shrink oversized pages' and 'Fit to printable area'. On a phone, choose 'Original size' / 'Do not scale'.
3. Print this page only. Measure the square with a steel ruler: both sides must be 100 mm, and the ruler marks must match your ruler. 1 mm off is fine; more is not.
4. If it is wrong, fix the printer setting and print this page again. Do not go on.
5. When it is right, print all the other pages with the SAME settings.
6. Big parts are split across pages (tiles A1, A2, B1...). Cut each tile on its border, lay the next tile over the grey overlap strip, line up the grey 50 mm grid and the measurement ruler, and tape it. Check the long measurement with a tape before cutting.
7. Glue-stick the paper on the thermocol, cut along the solid black line, cut out the red parts, and push a skewer through every + mark.

Line types:

- cut line
- cut out (window / pocket)
- fold / glue line
- placement guide, do not cut
- skewer hole: lines up the layers
- measurement (mm), do no

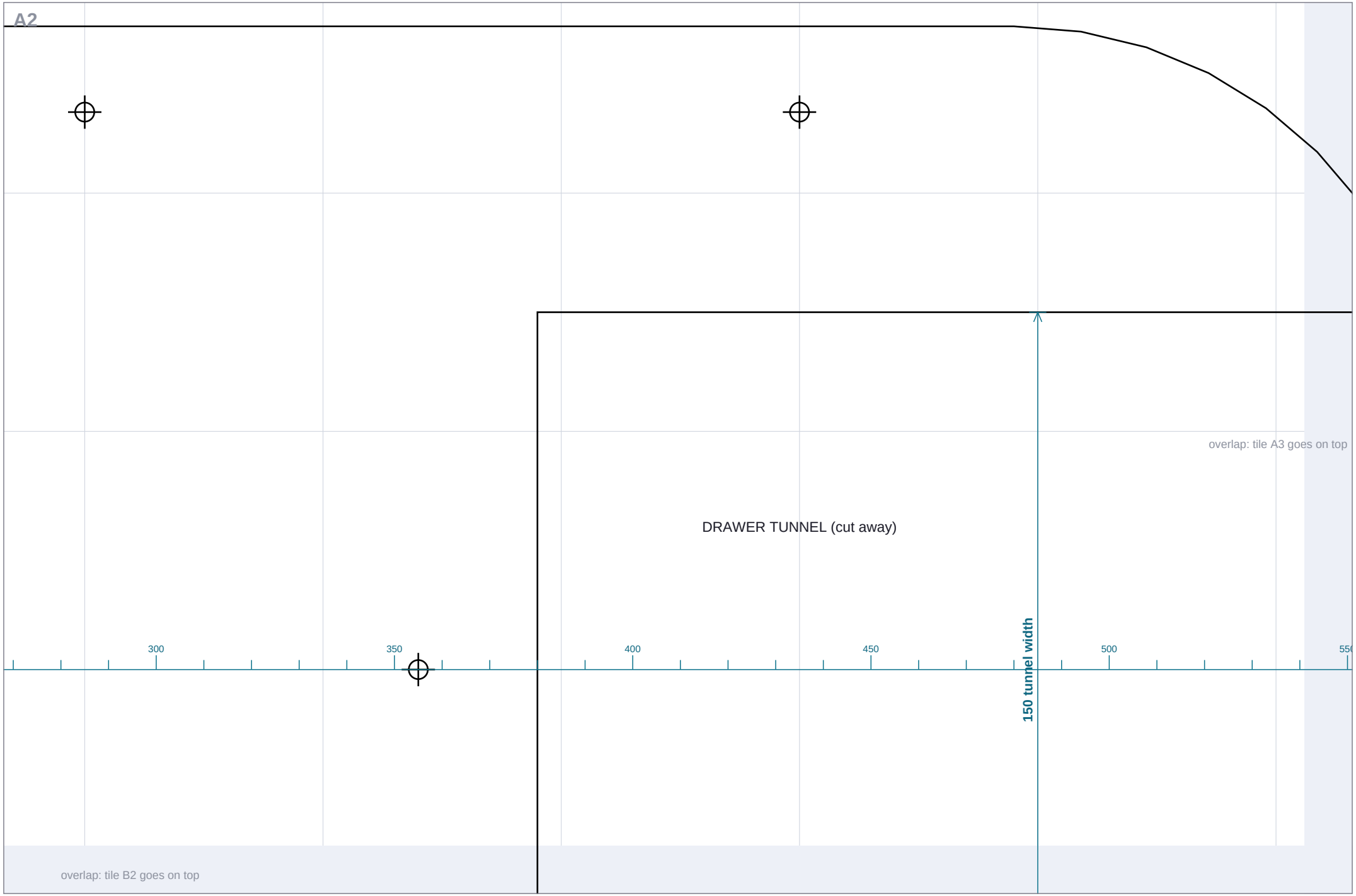
VARUNA AI Pod | Base layer 1 - tile A1 of B3 (25 mm thermocol)

<- must measure 50 mm



VARUNA AI Pod | Base layer 1 - tile A2 of B3 (25 mm thermocol)

<- must measure 50 mm



VARUNA AI Pod | Base layer 1 - tile A3 of B3 (25 mm thermocol)

<- must measure 50 mm

A3

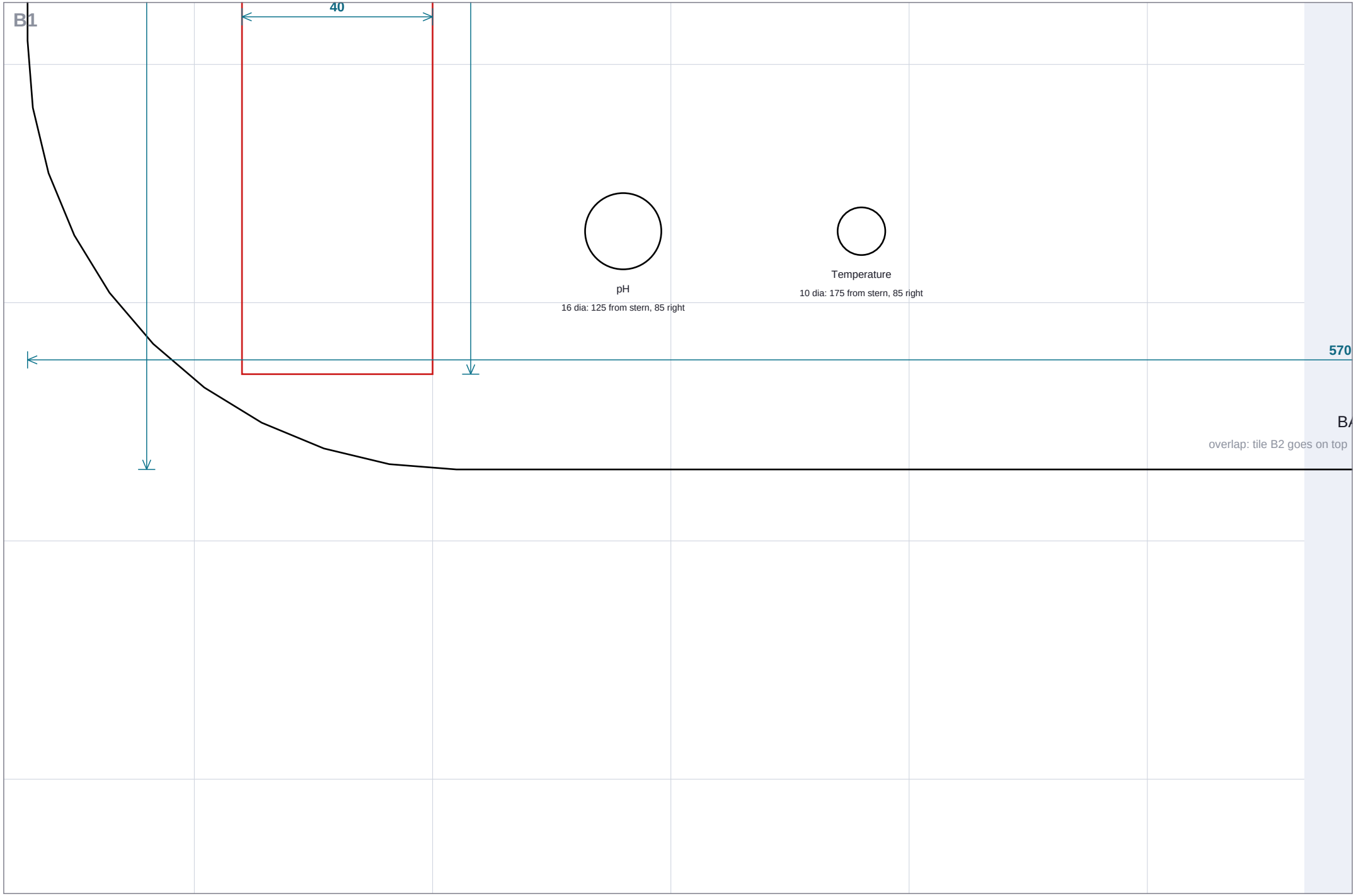




overlap: tile B3 goes on top

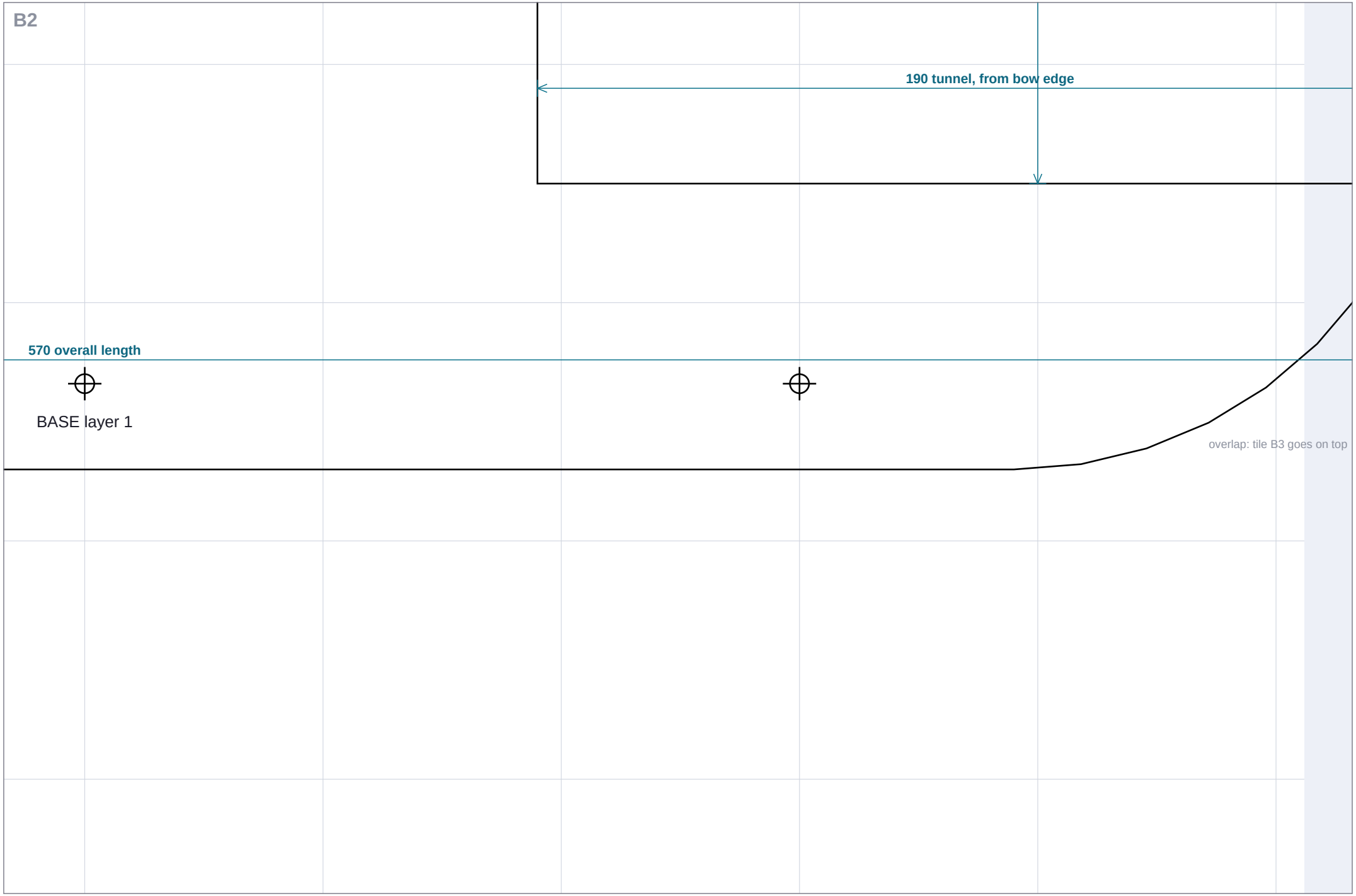
VARUNA AI Pod | Base layer 1 - tile B1 of B3 (25 mm thermocol)

<- must measure 50 mm



VARUNA AI Pod | Base layer 1 - tile B2 of B3 (25 mm thermocol)

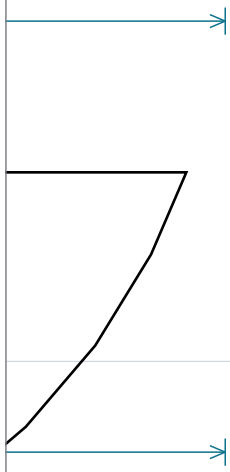
<- must measure 50 mm



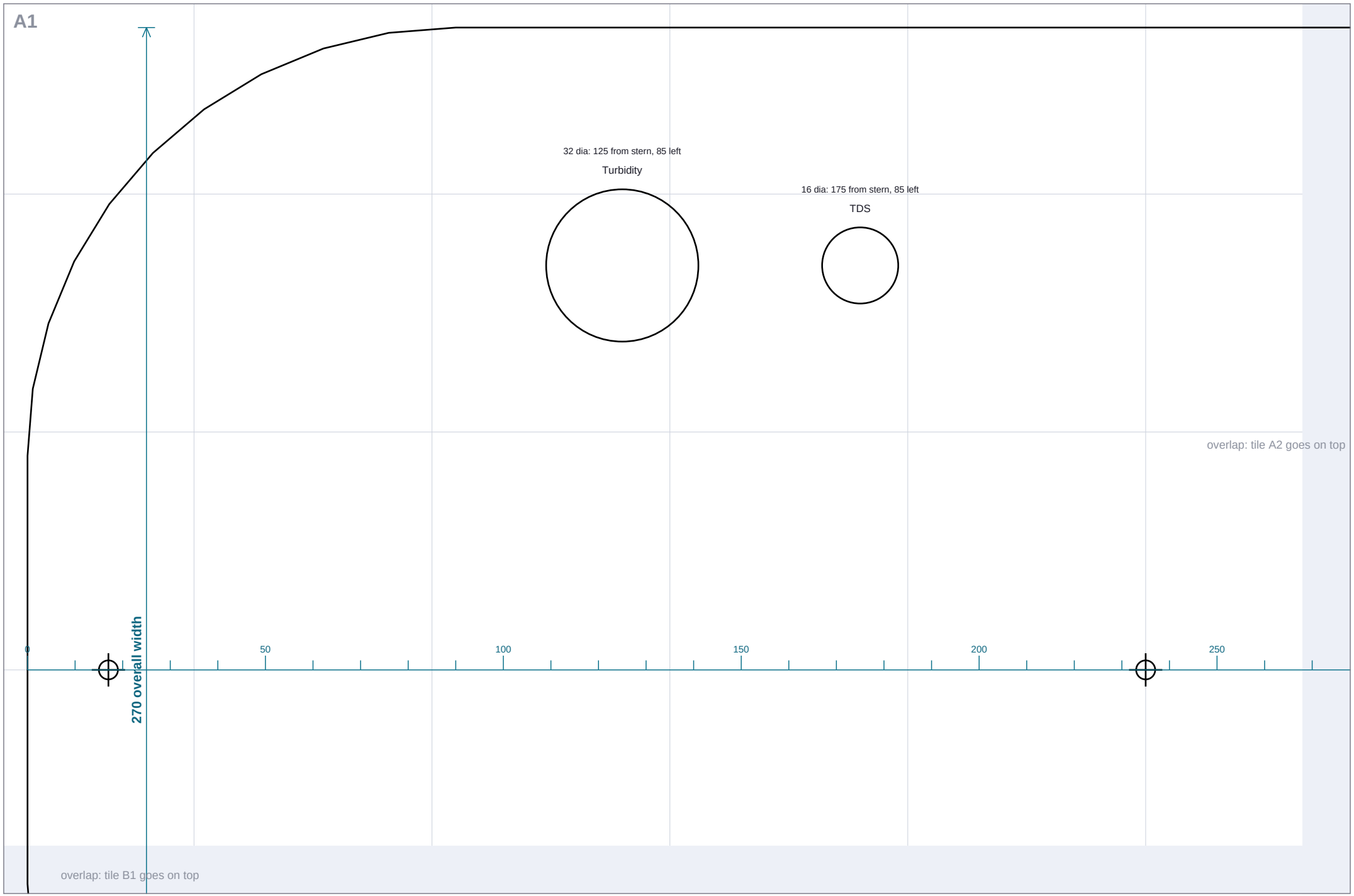
VARUNA AI Pod | Base layer 1 - tile B3 of B3 (25 mm thermocol)

<- must measure 50 mm

B3

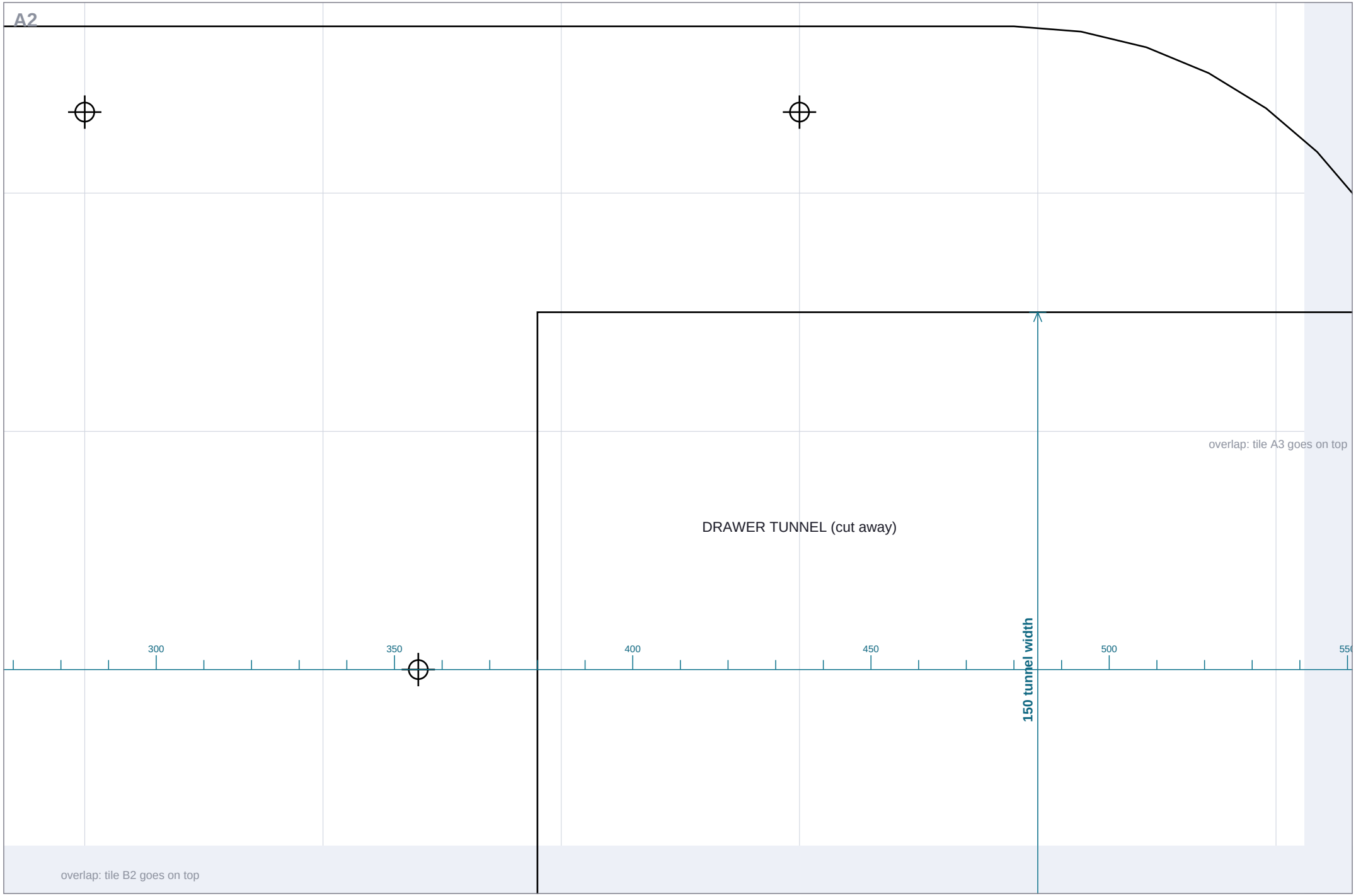


<- must measure 50 mm



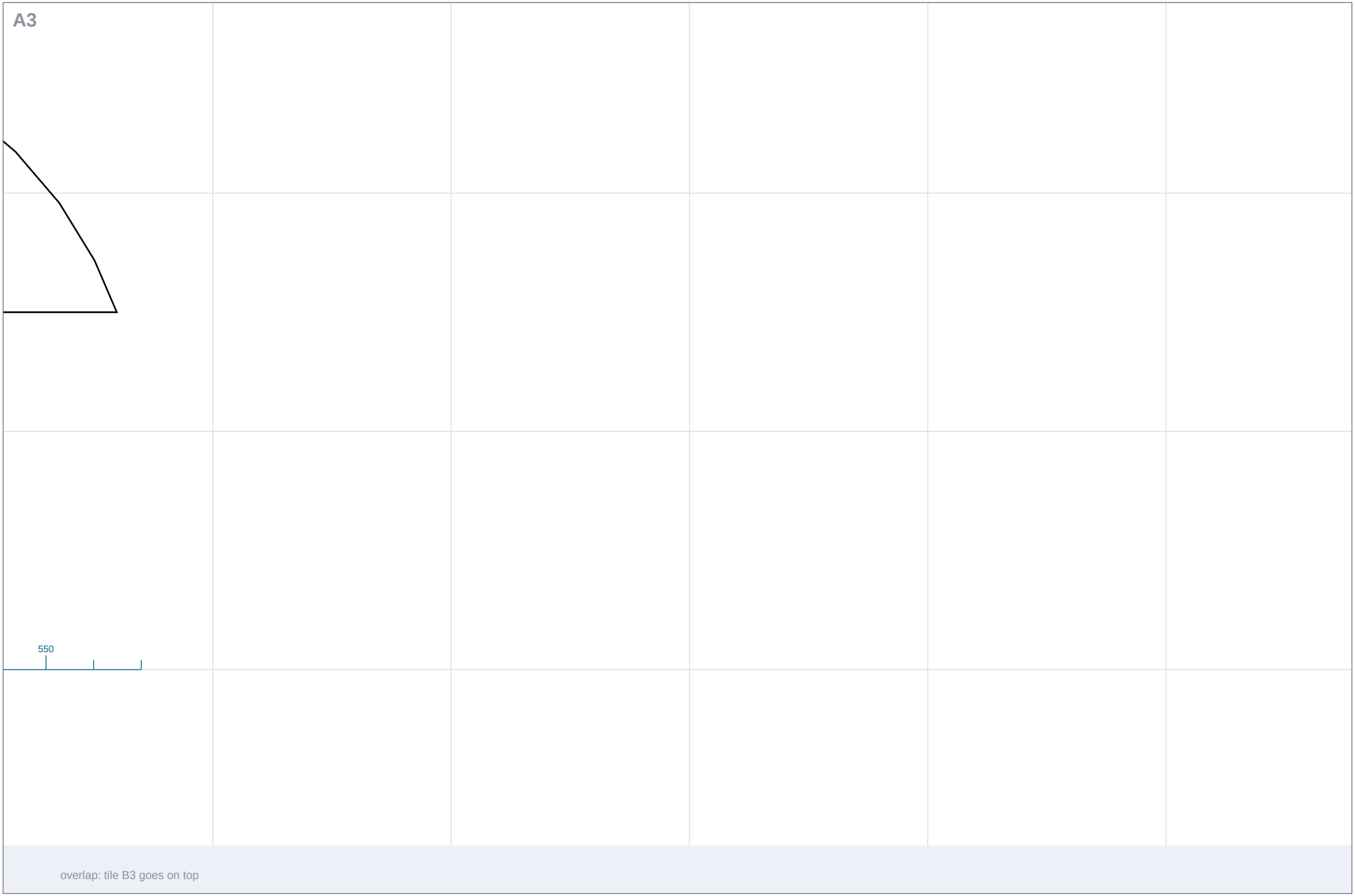
VARUNA AI Pod | Base layer 2 - tile A2 of B3 (25 mm thermocol)

<- must measure 50 mm



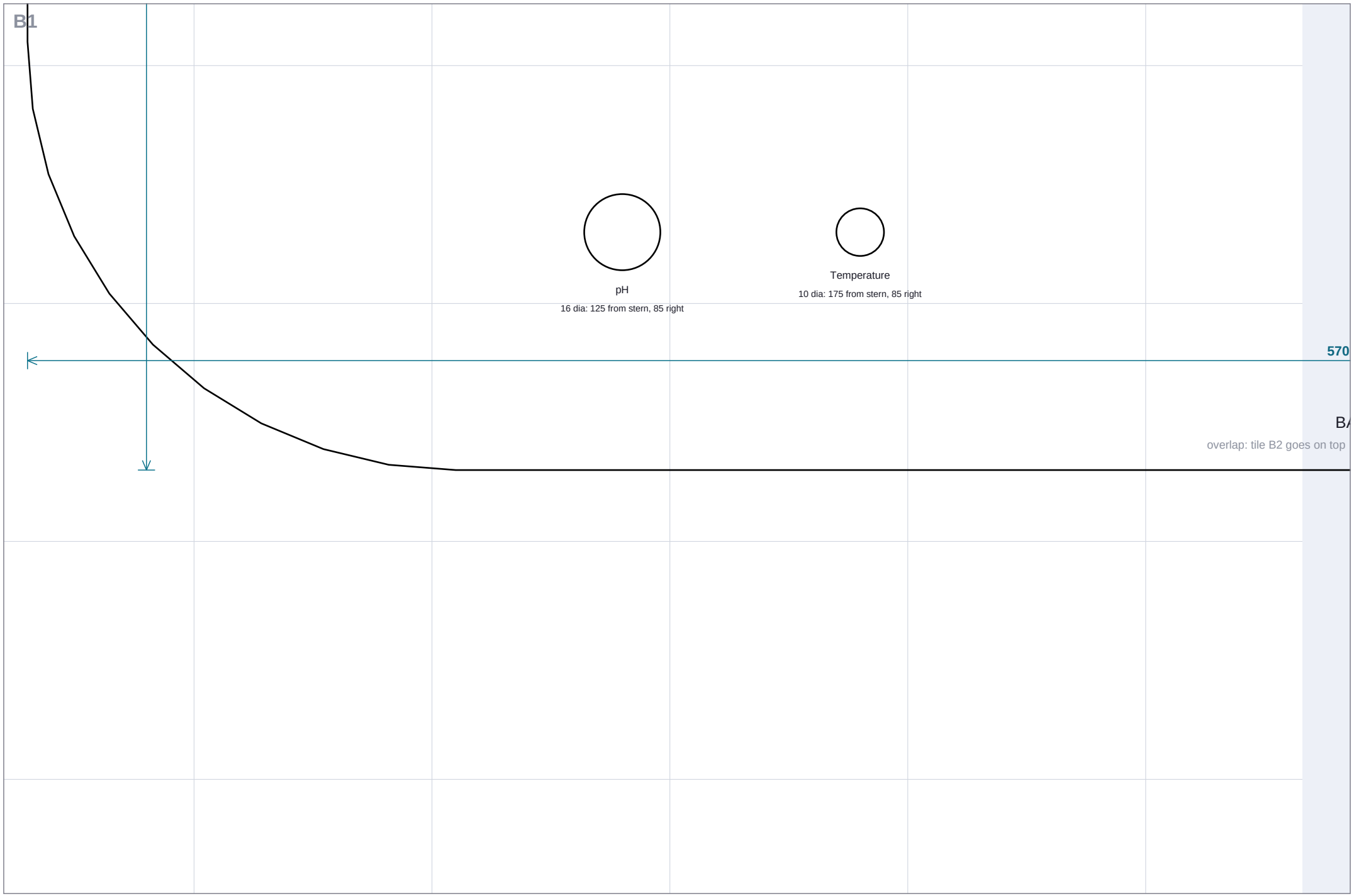
VARUNA AI Pod | Base layer 2 - tile A3 of B3 (25 mm thermocol)

<- must measure 50 mm

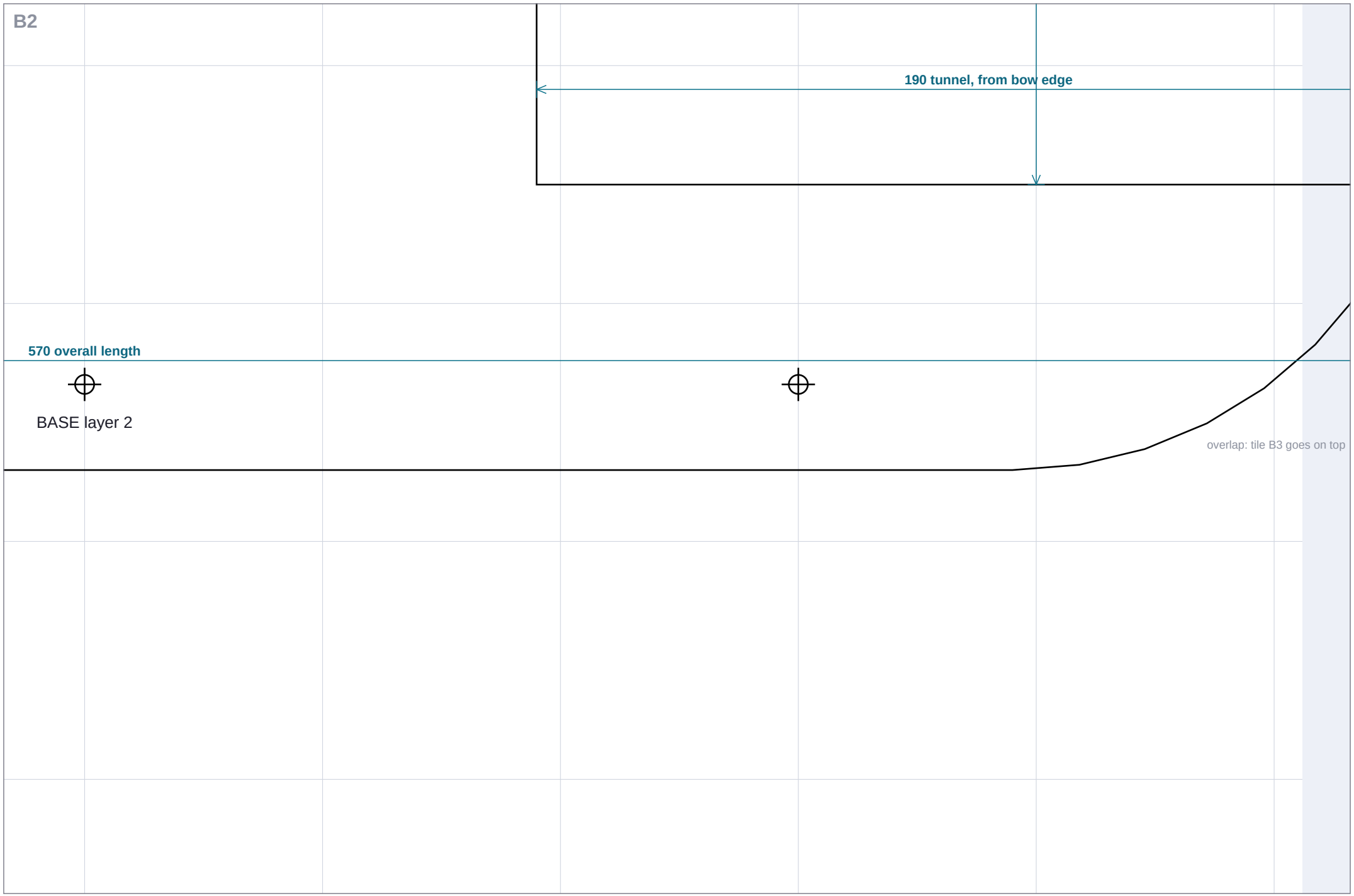


VARUNA AI Pod | Base layer 2 - tile B1 of B3 (25 mm thermocol)

<- must measure 50 mm



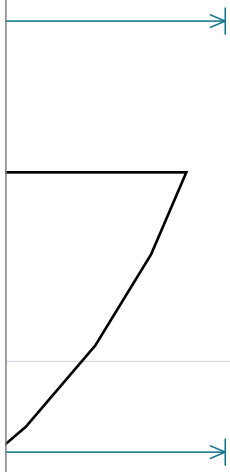
VARUNA AI Pod | Base layer 2 - tile B2 of B3 (25 mm thermocol)



VARUNA AI Pod | Base layer 2 - tile B3 of B3 (25 mm thermocol)

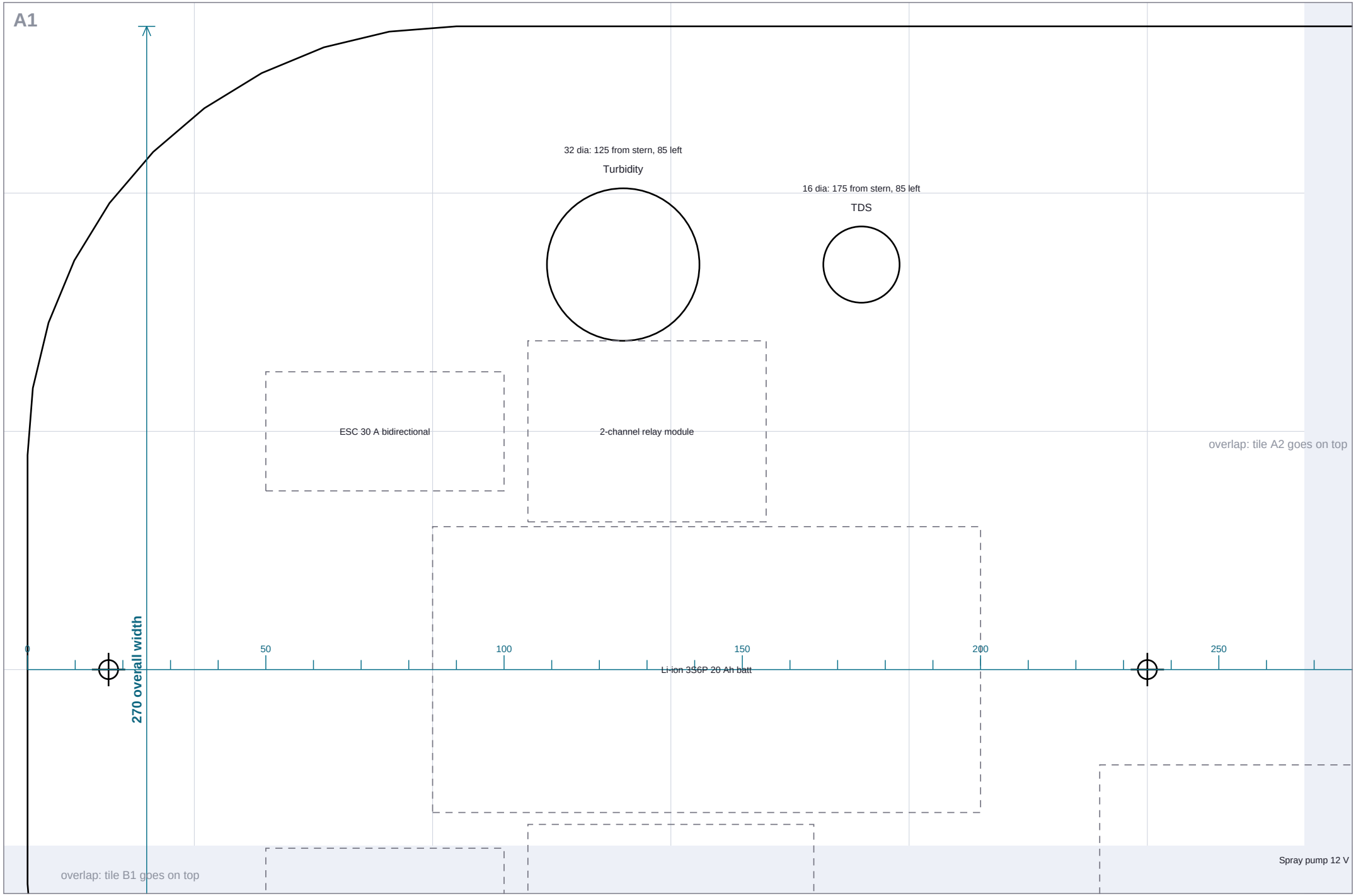
<- must measure 50 mm

B3



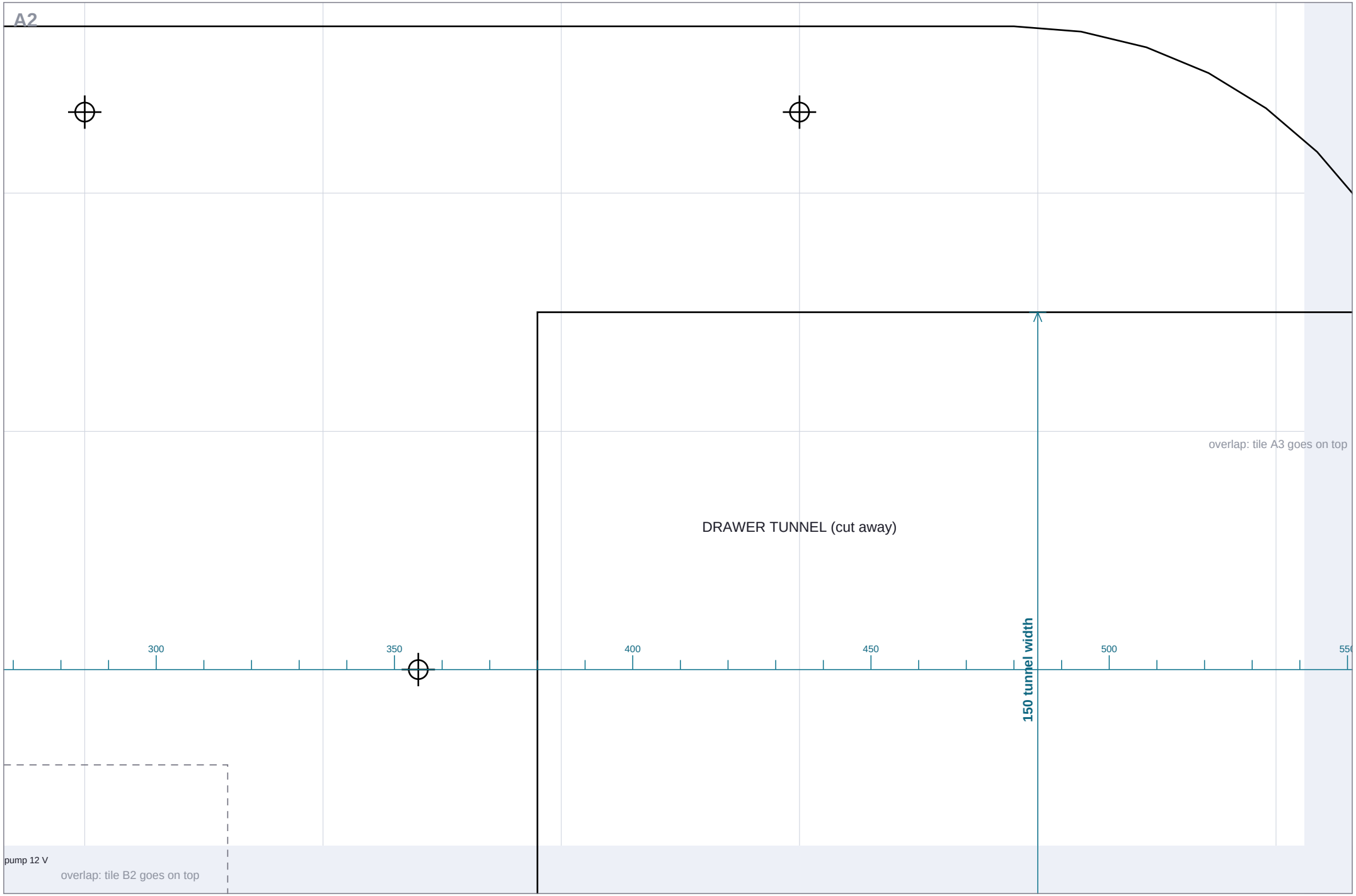
VARUNA AI Pod | Base layer 3 - tile A1 of B3 (25 mm thermocol)

<- must measure 50 mm



VARUNA AI Pod | Base layer 3 - tile A2 of B3 (25 mm thermocol)

<- must measure 50 mm



VARUNA AI Pod | Base layer 3 - tile A3 of B3 (25 mm thermocol)

<- must measure 50 mm

A3

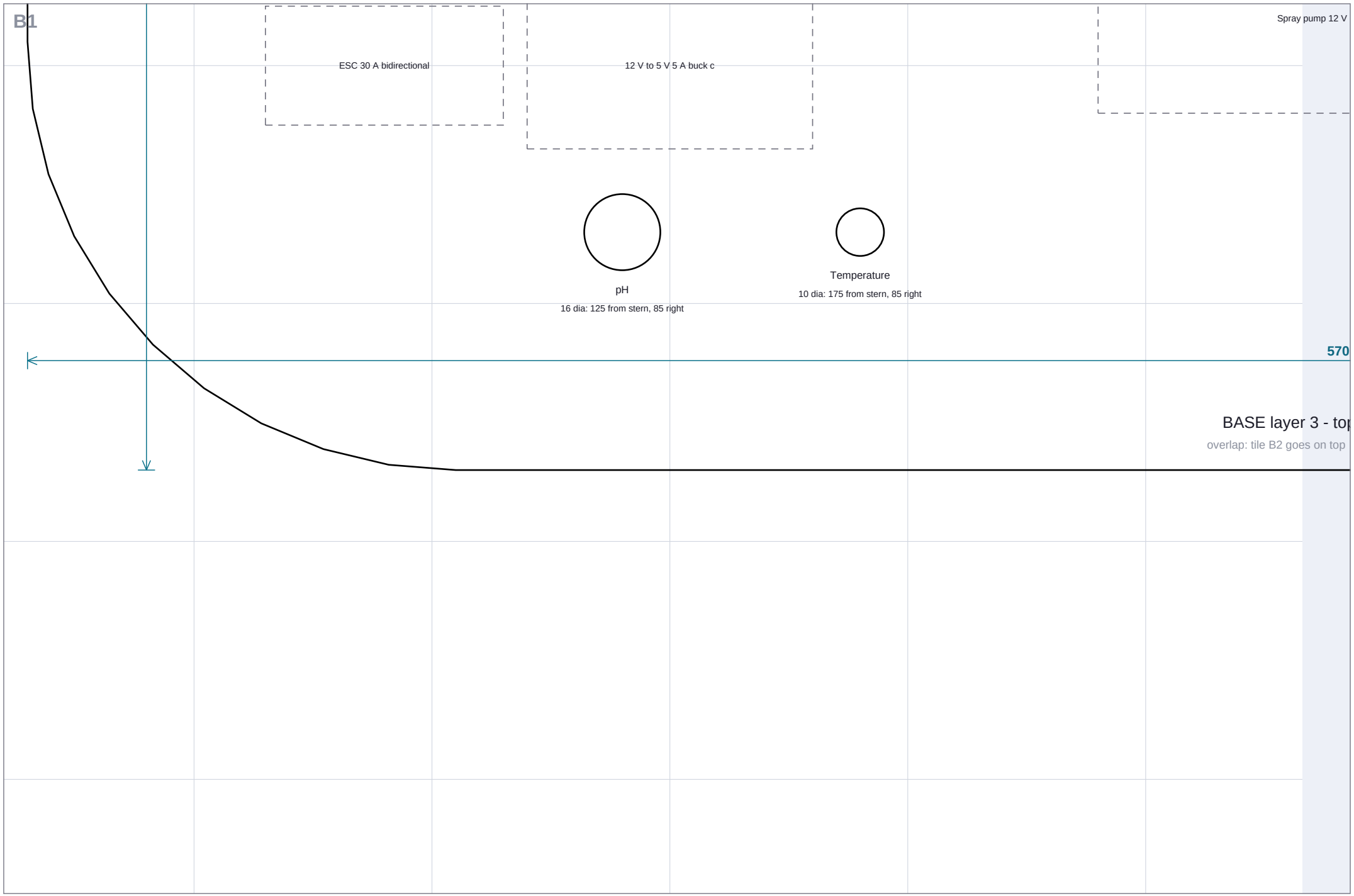




overlap: tile B3 goes on top

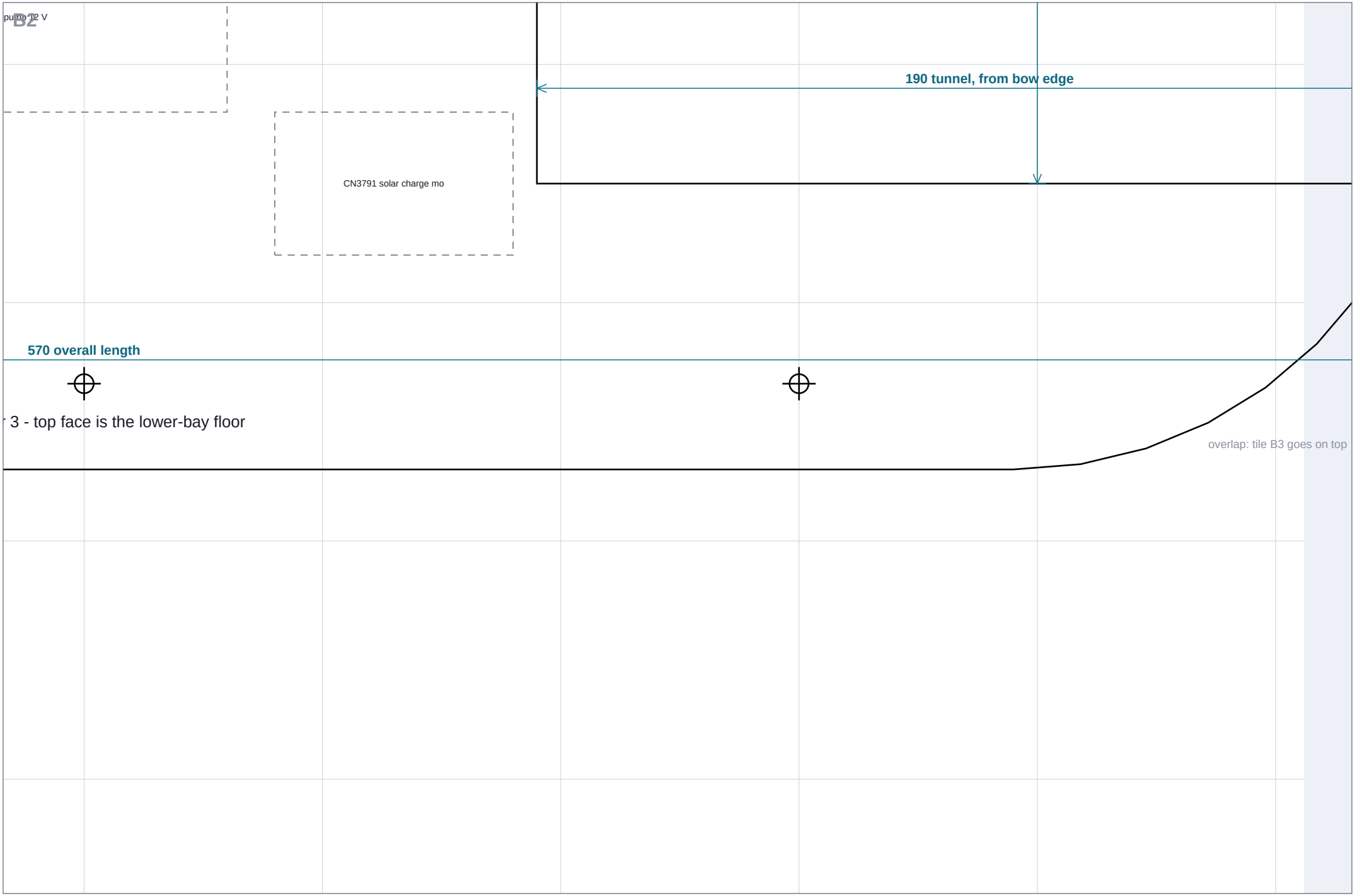
VARUNA AI Pod | Base layer 3 - tile B1 of B3 (25 mm thermocol)

<- must measure 50 mm



VARUNA AI Pod | Base layer 3 - tile B2 of B3 (25 mm thermocol)

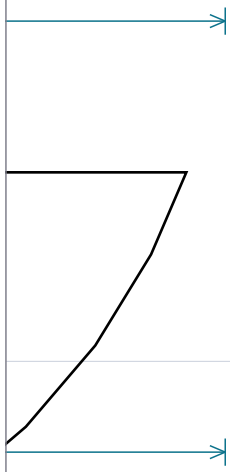
<- must measure 50 mm



VARUNA AI Pod | Base layer 3 - tile B3 of B3 (25 mm thermocol)

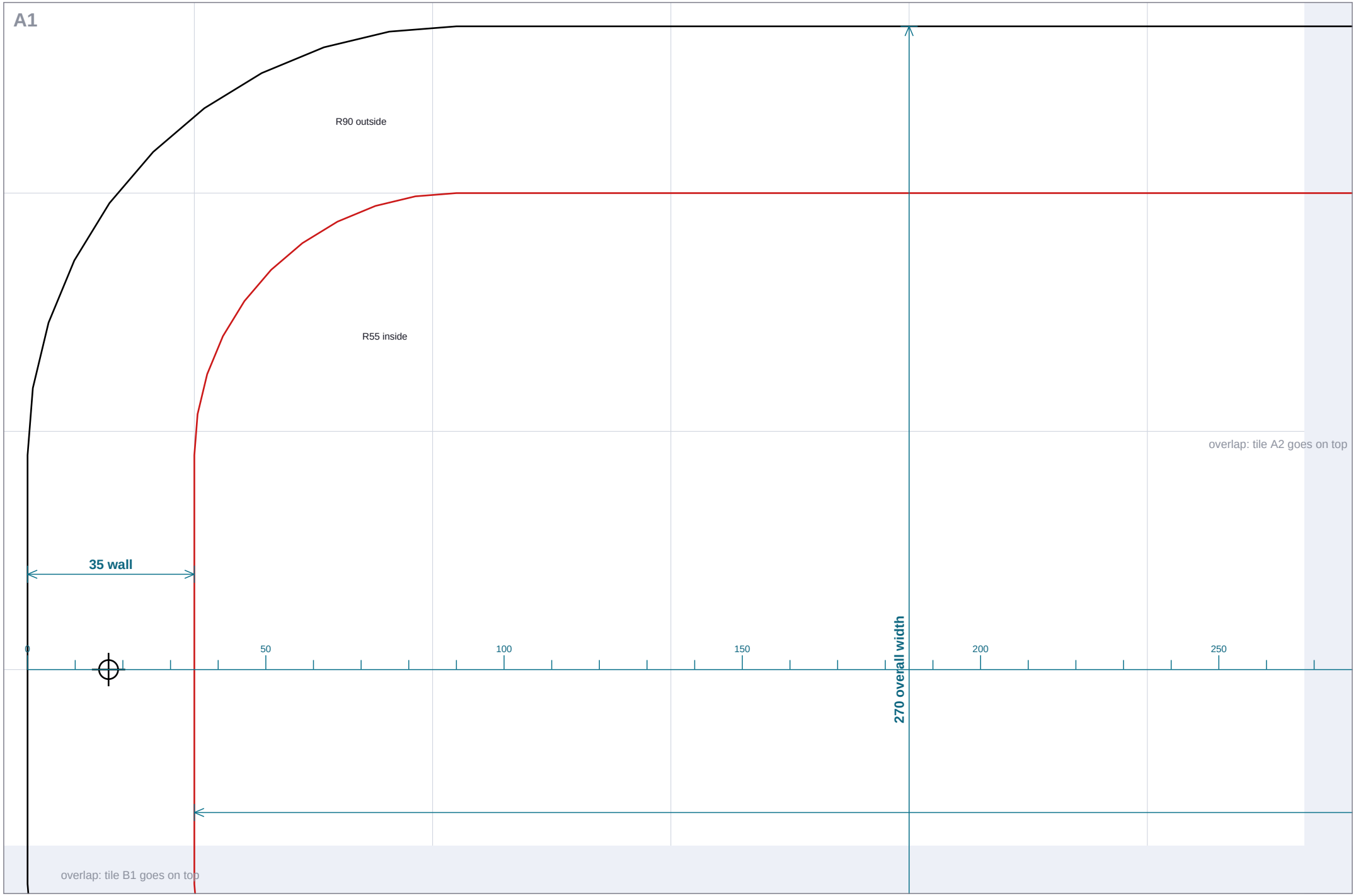
<- must measure 50 mm

B3

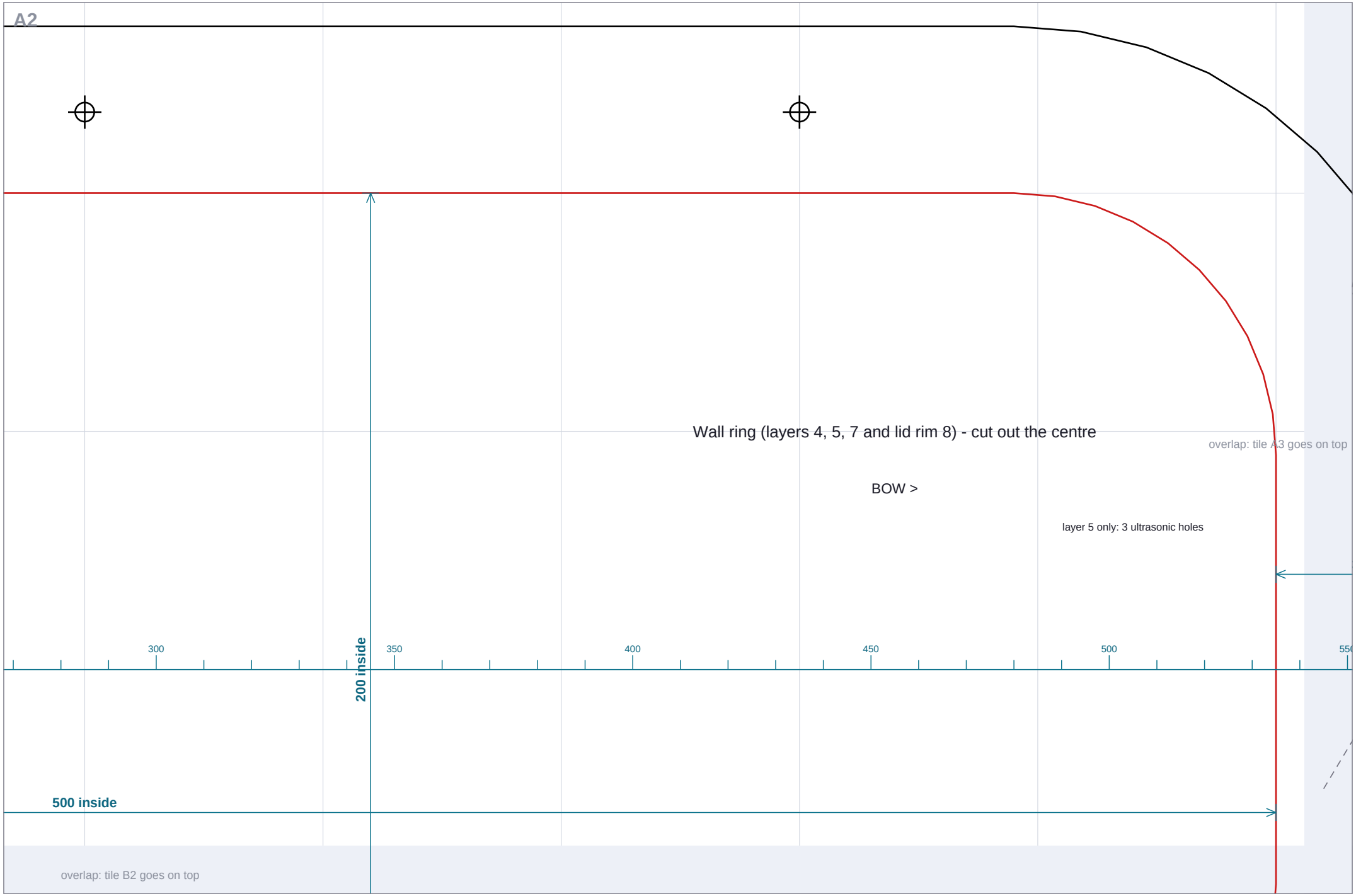


VARUNA AI Pod | Wall ring (layers 4, 5, 7 and lid rim 8) - tile A1 of B3 (25 mm thermocol) PRINT 4 COPIES

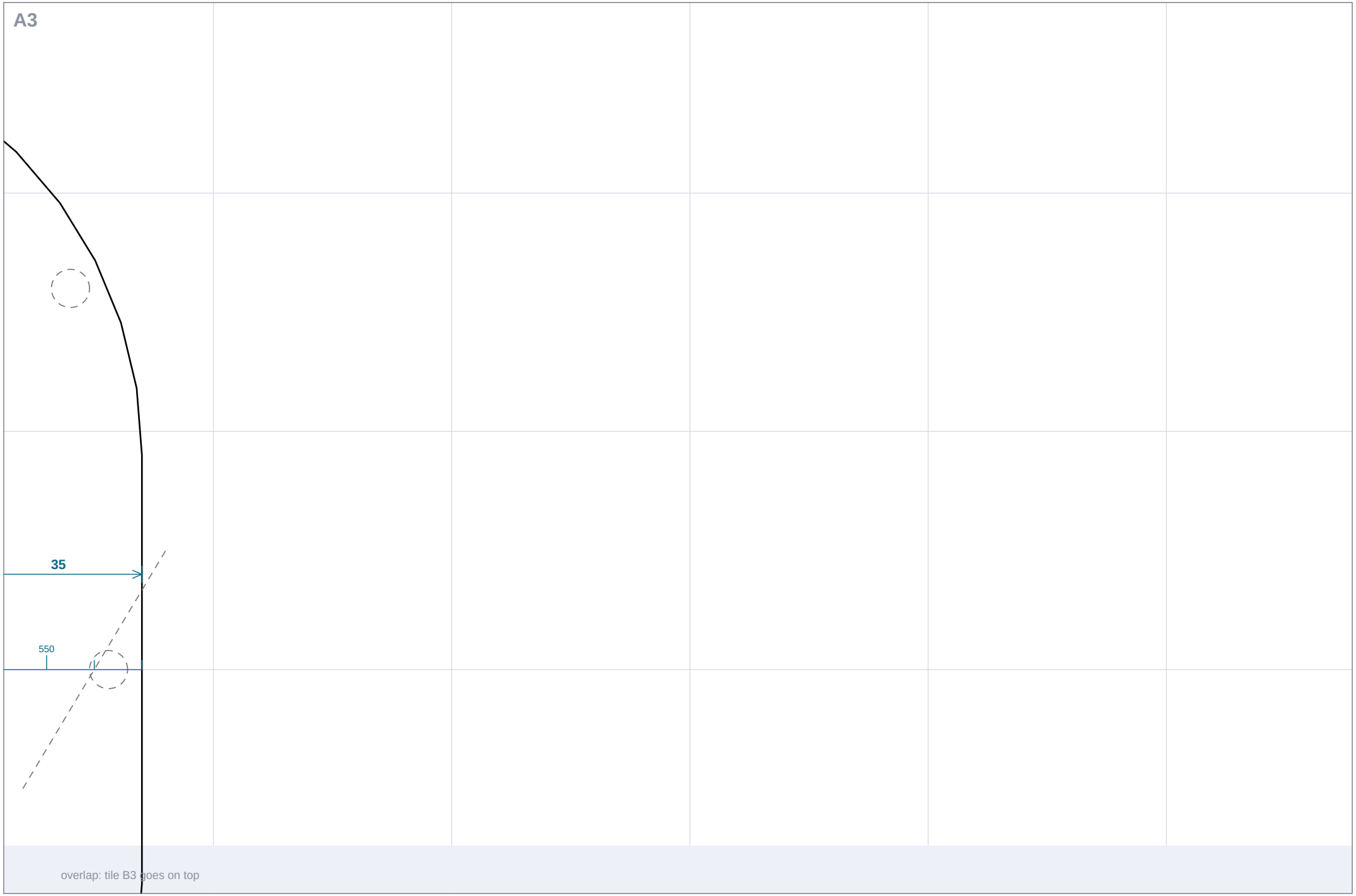
<- must measure 50 mm



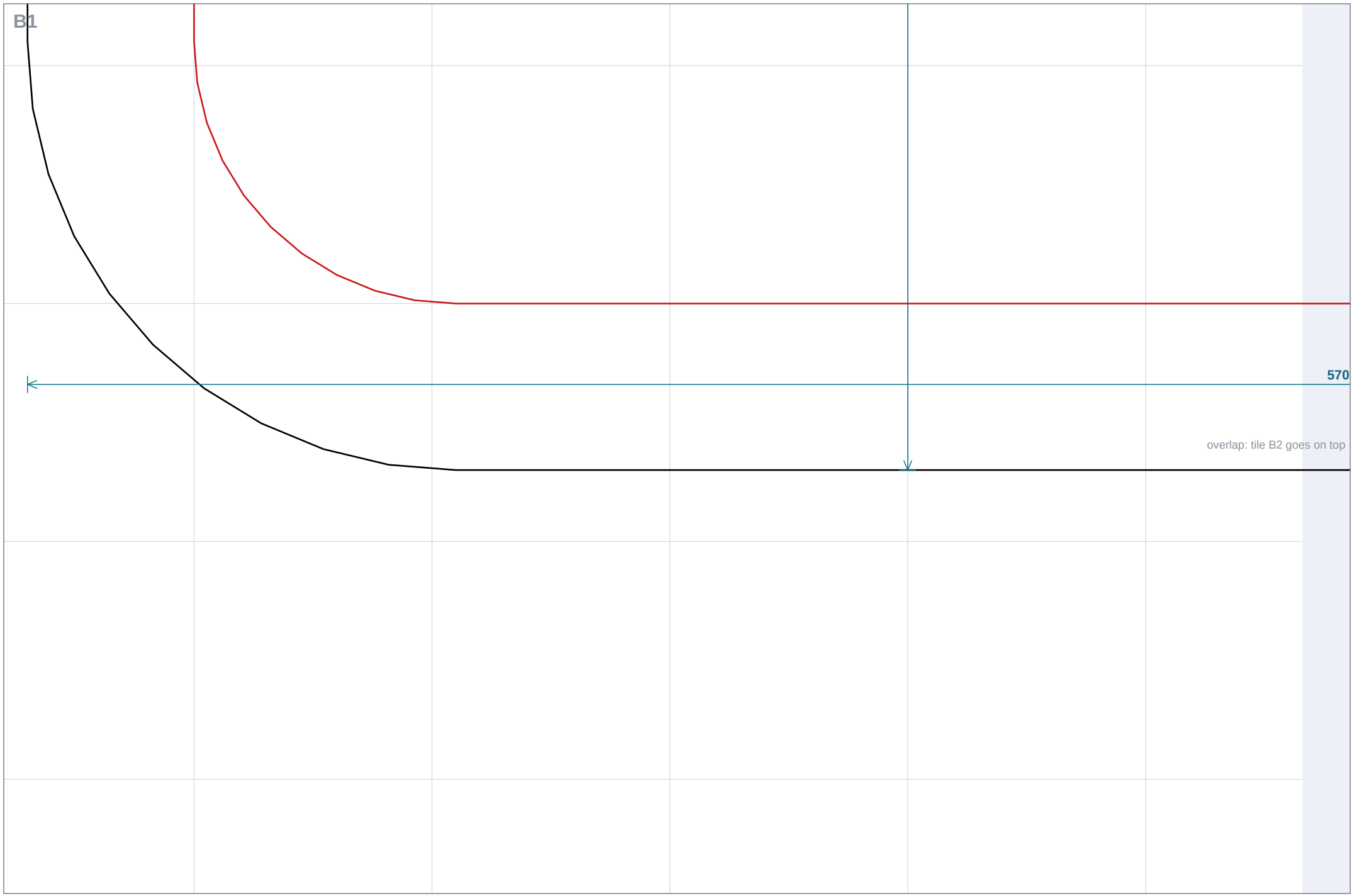
<- must measure 50 mm



<- must measure 50 mm

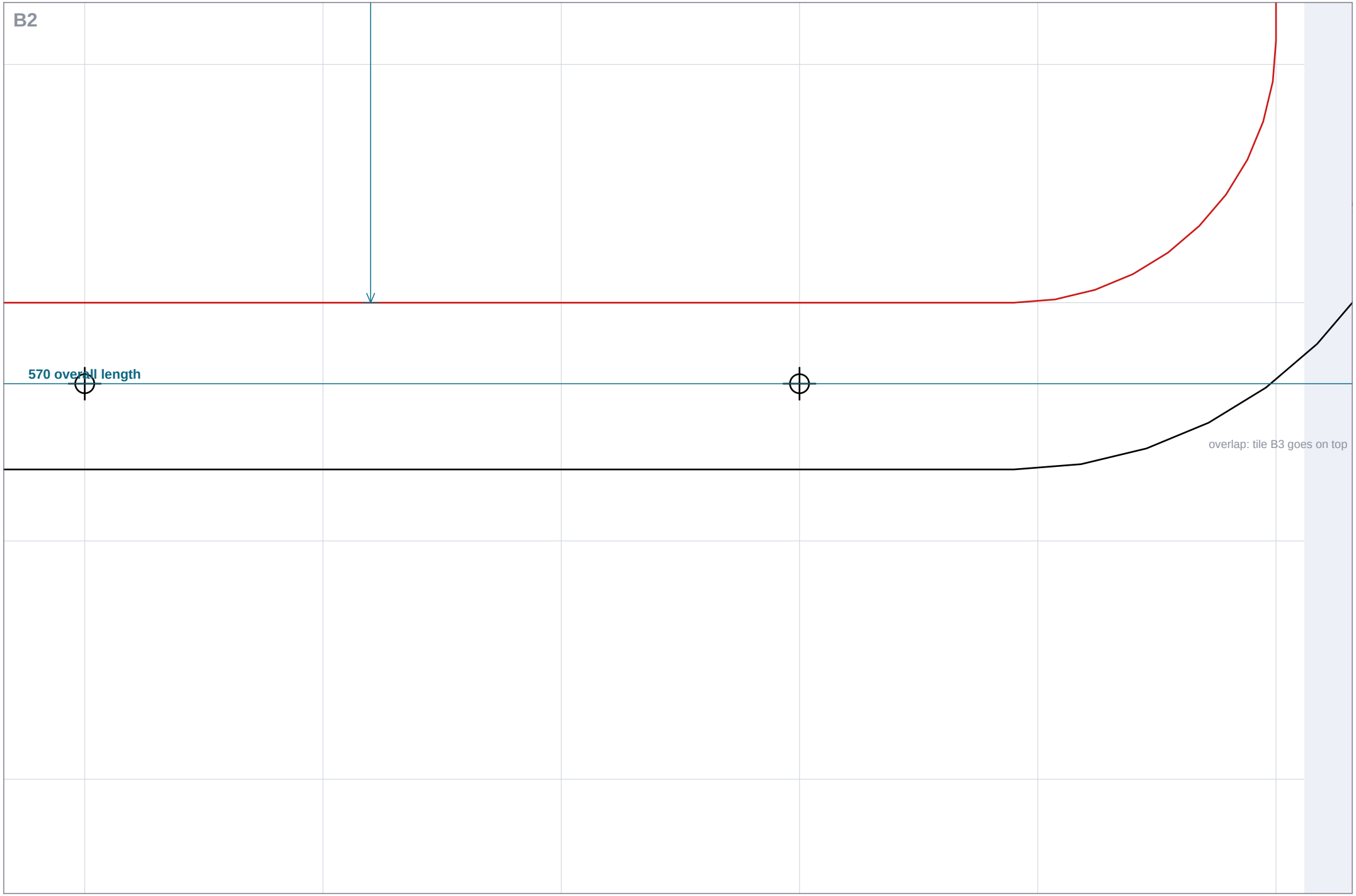


<- must measure 50 mm



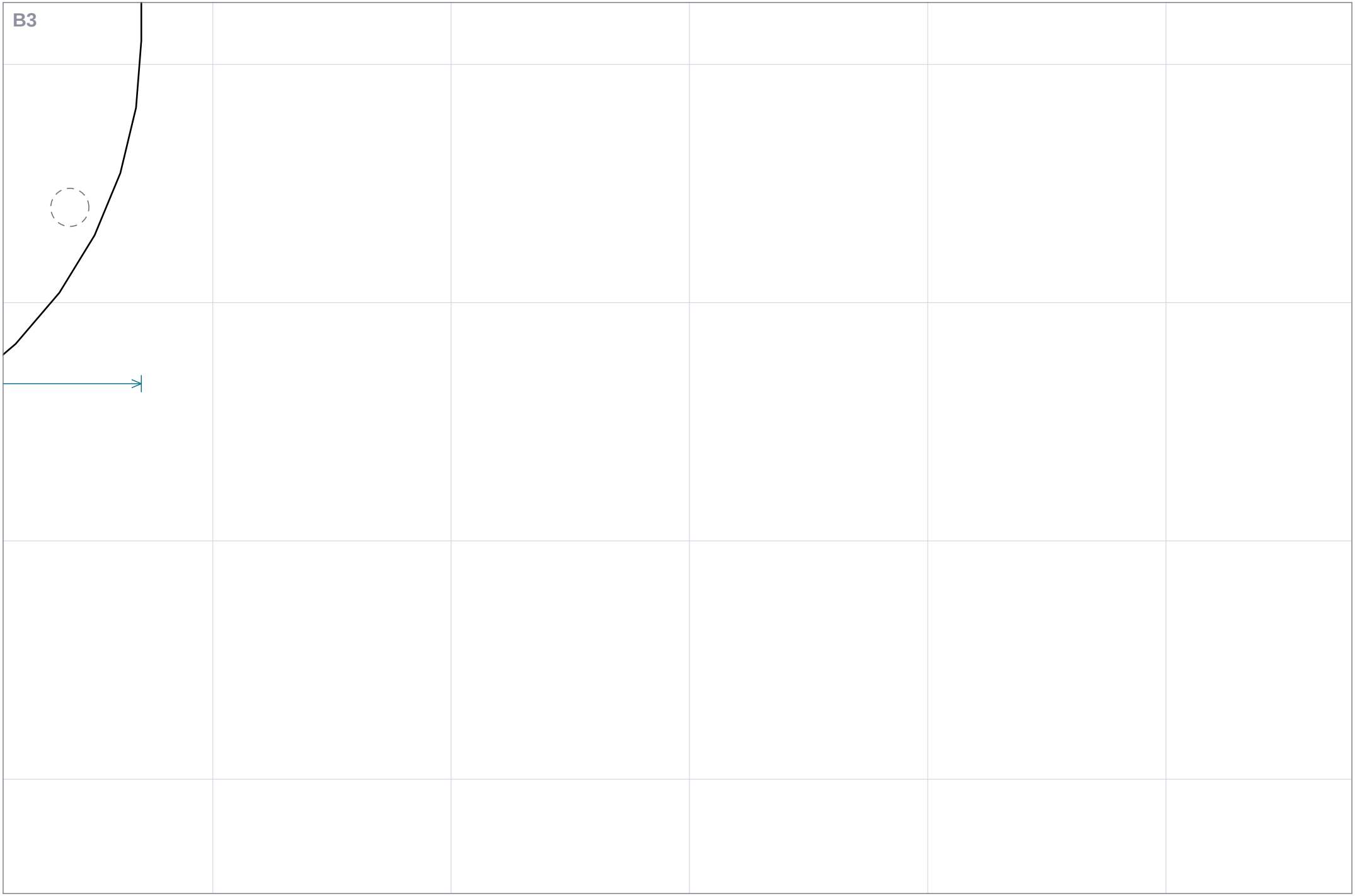
VARUNA AI Pod | Wall ring (layers 4, 5, 7 and lid rim 8) - tile B2 of B3 (25 mm thermocol) PRINT 4 COPIES

<- must measure 50 mm



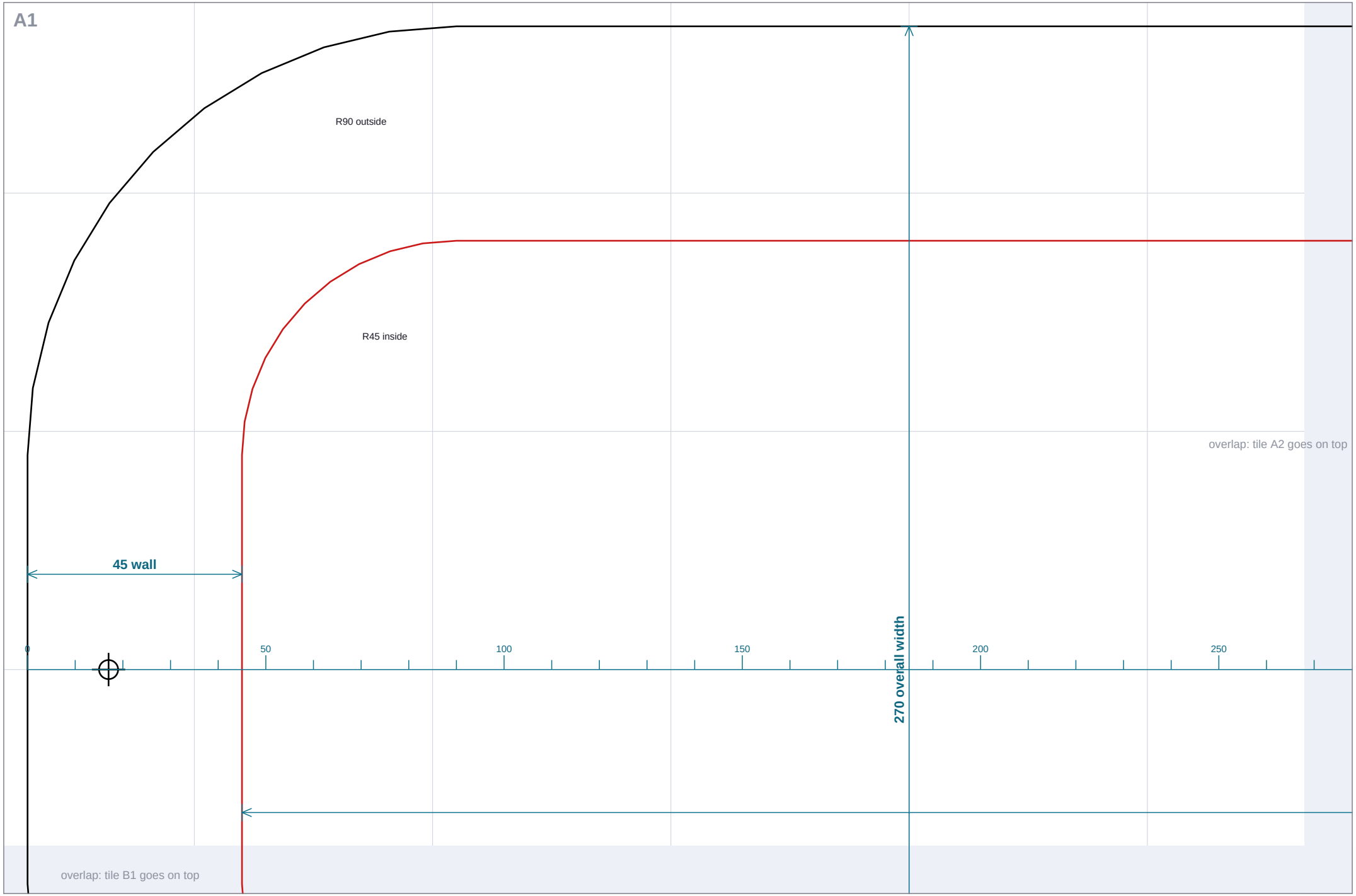
VARUNA AI Pod | Wall ring (layers 4, 5, 7 and lid rim 8) - tile B3 of B3 (25 mm thermocol) PRINT 4 COPIES

<- must measure 50 mm



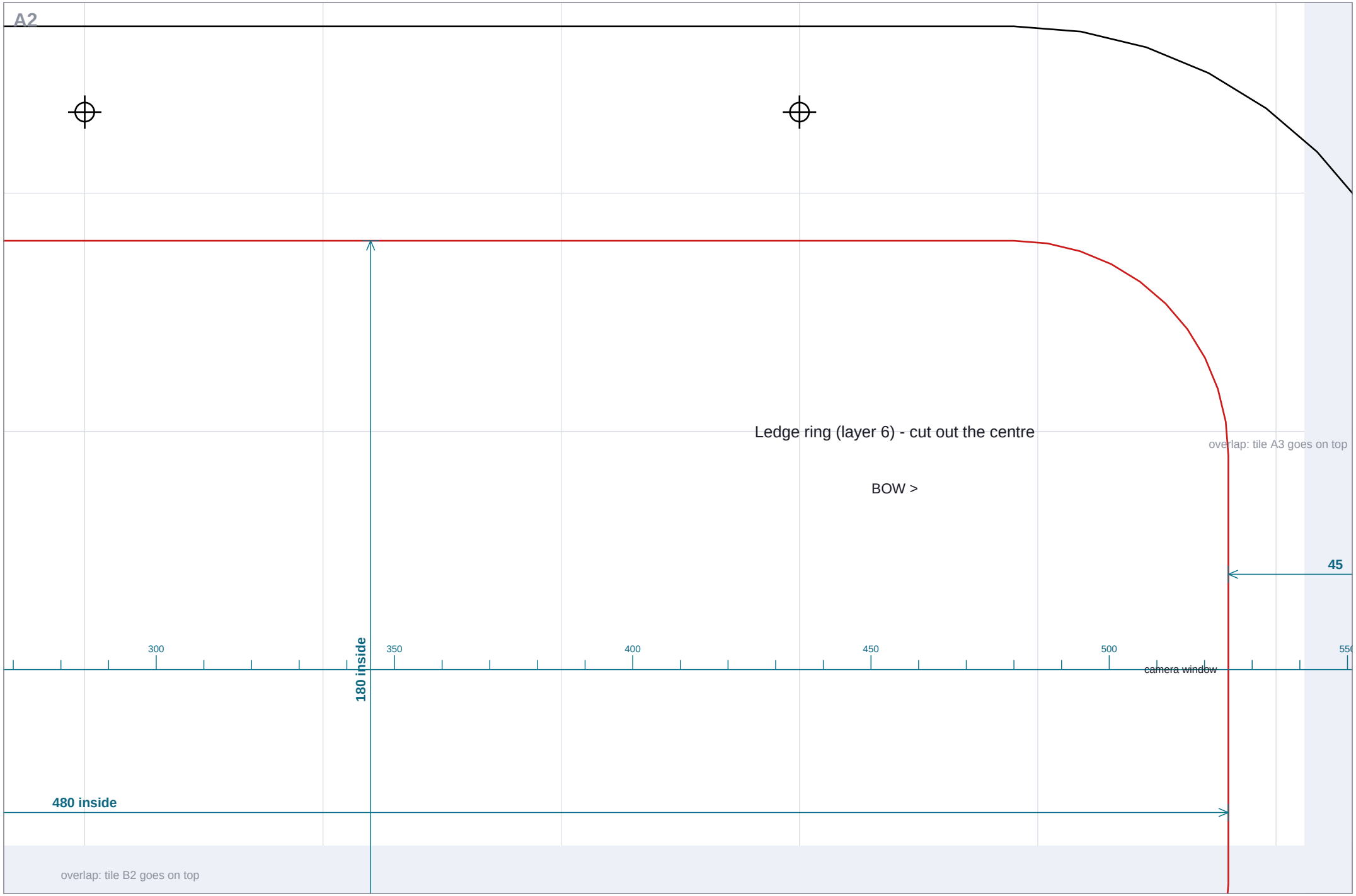
VARUNA AI Pod | Ledge ring (layer 6) - tile A1 of B3 (25 mm thermocol)

<- must measure 50 mm



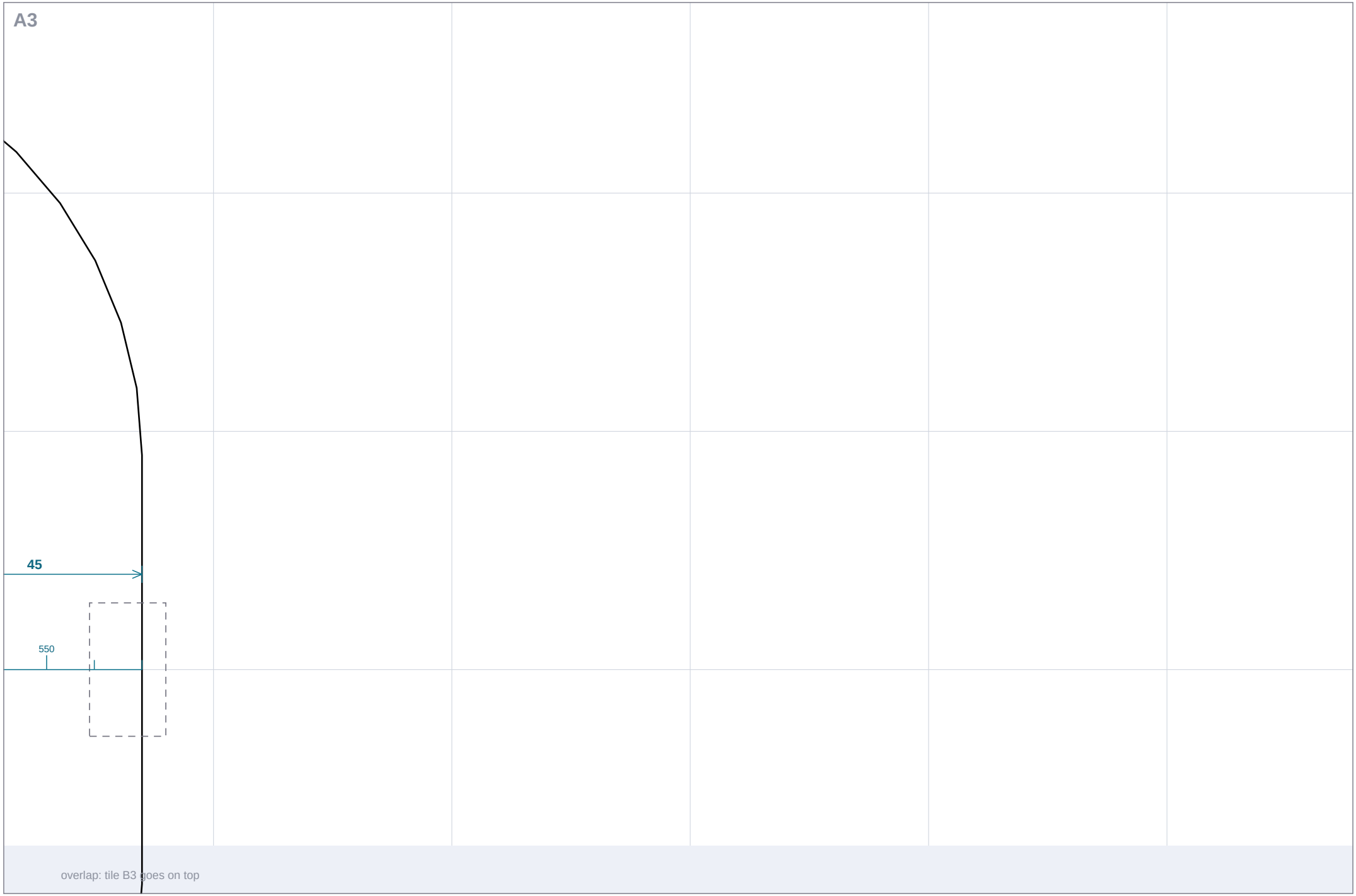
VARUNA AI Pod | Ledge ring (layer 6) - tile A2 of B3 (25 mm thermocol)

<- must measure 50 mm



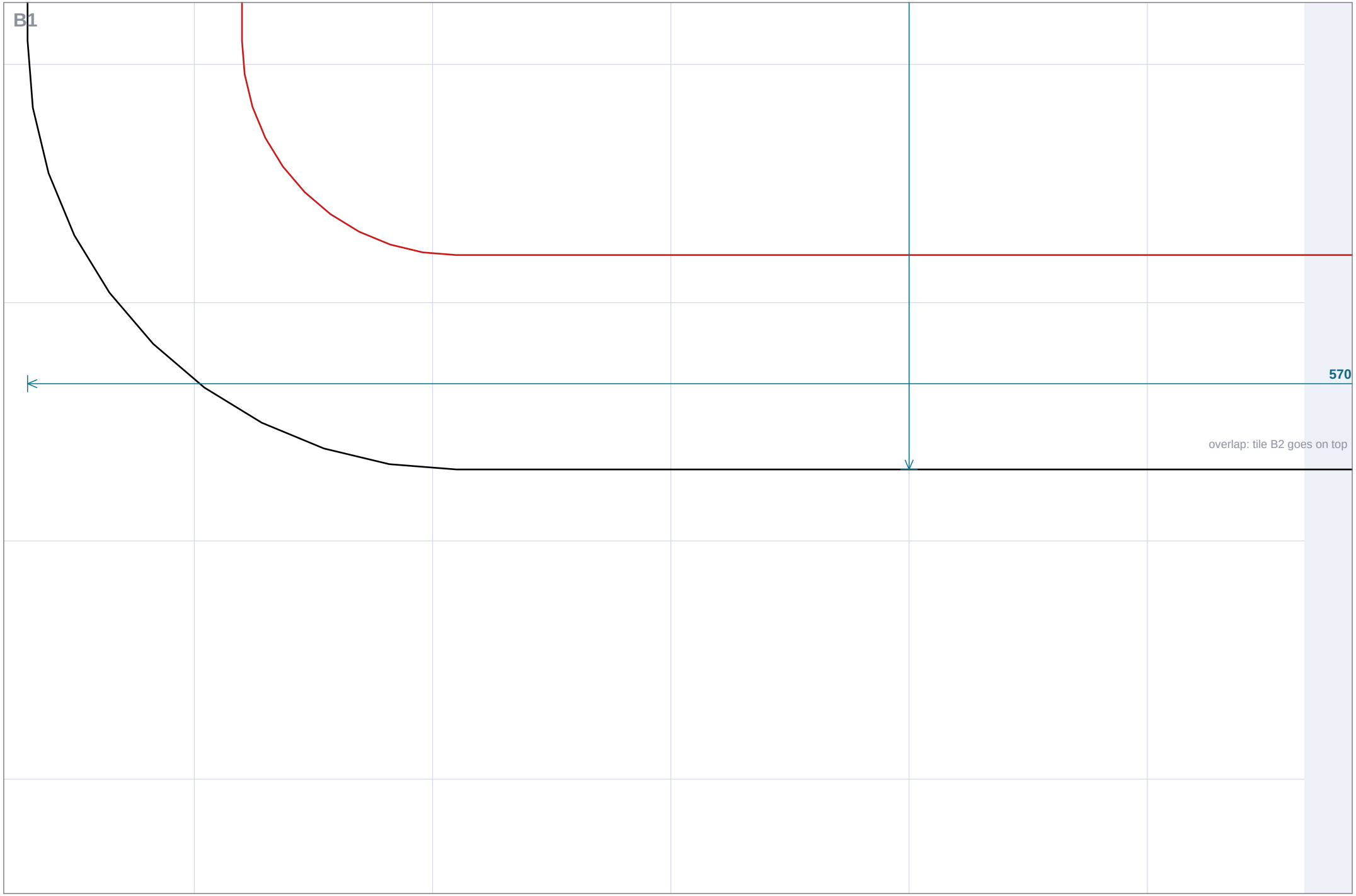
VARUNA AI Pod | Ledge ring (layer 6) - tile A3 of B3 (25 mm thermocol)

<- must measure 50 mm



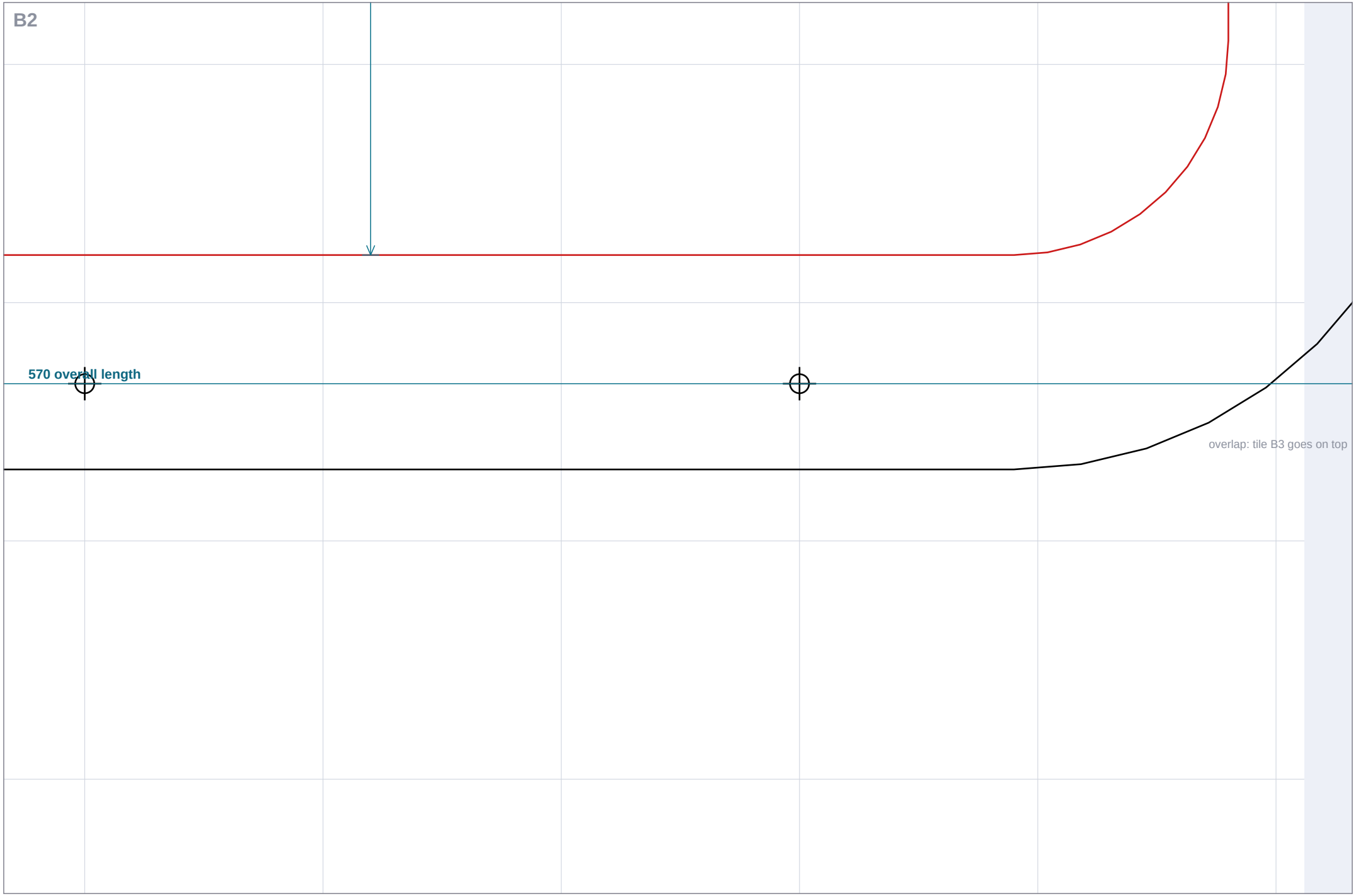
VARUNA AI Pod | Ledge ring (layer 6) - tile B1 of B3 (25 mm thermocol)

<- must measure 50 mm



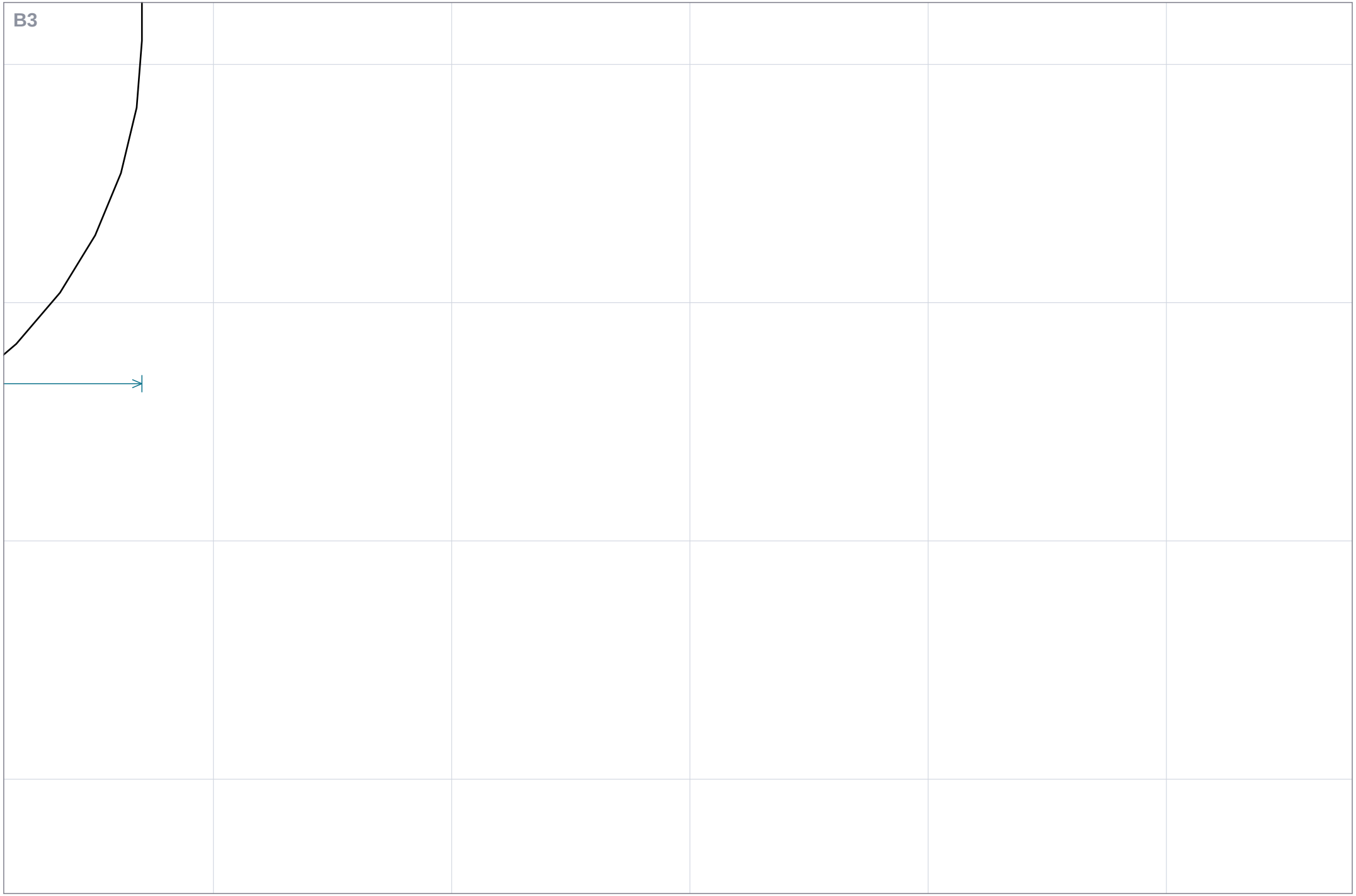
VARUNA AI Pod | Ledge ring (layer 6) - tile B2 of B3 (25 mm thermocol)

<- must measure 50 mm



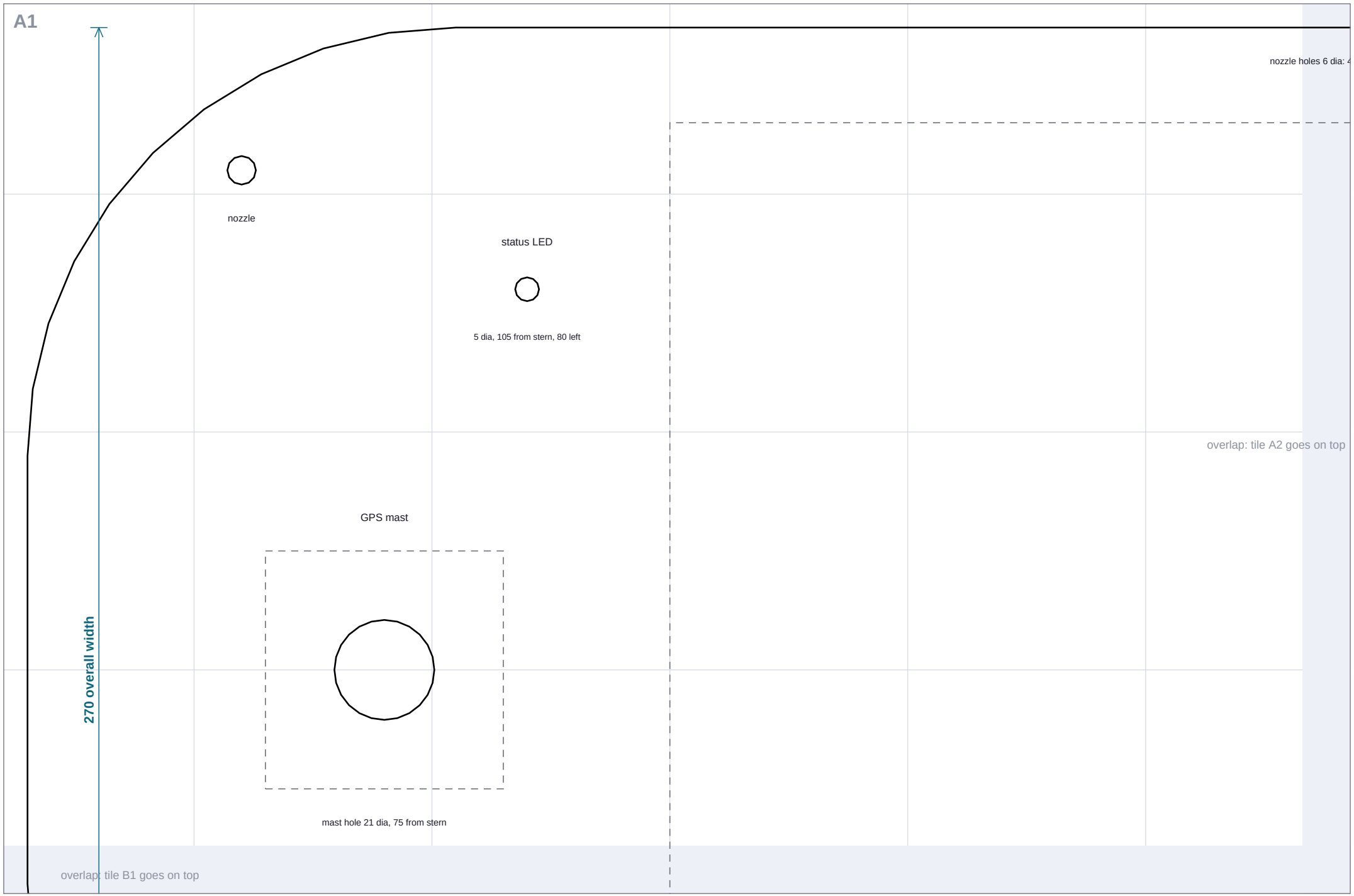
VARUNA AI Pod | Ledge ring (layer 6) - tile B3 of B3 (25 mm thermocol)

<- must measure 50 mm



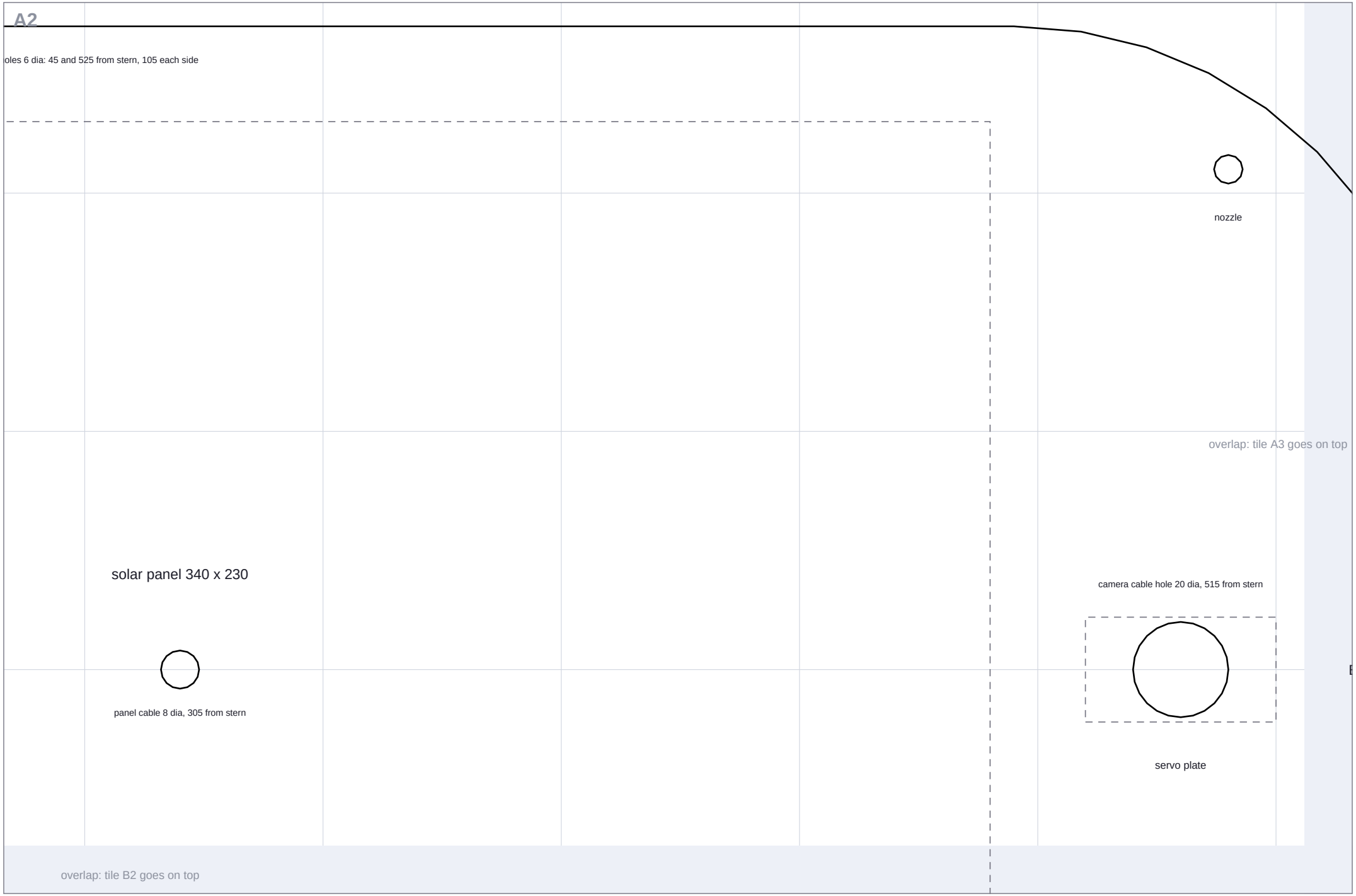
VARUNA AI Pod | Lid top - tile A1 of B3 (3 mm sunboard)

<- must measure 50 mm



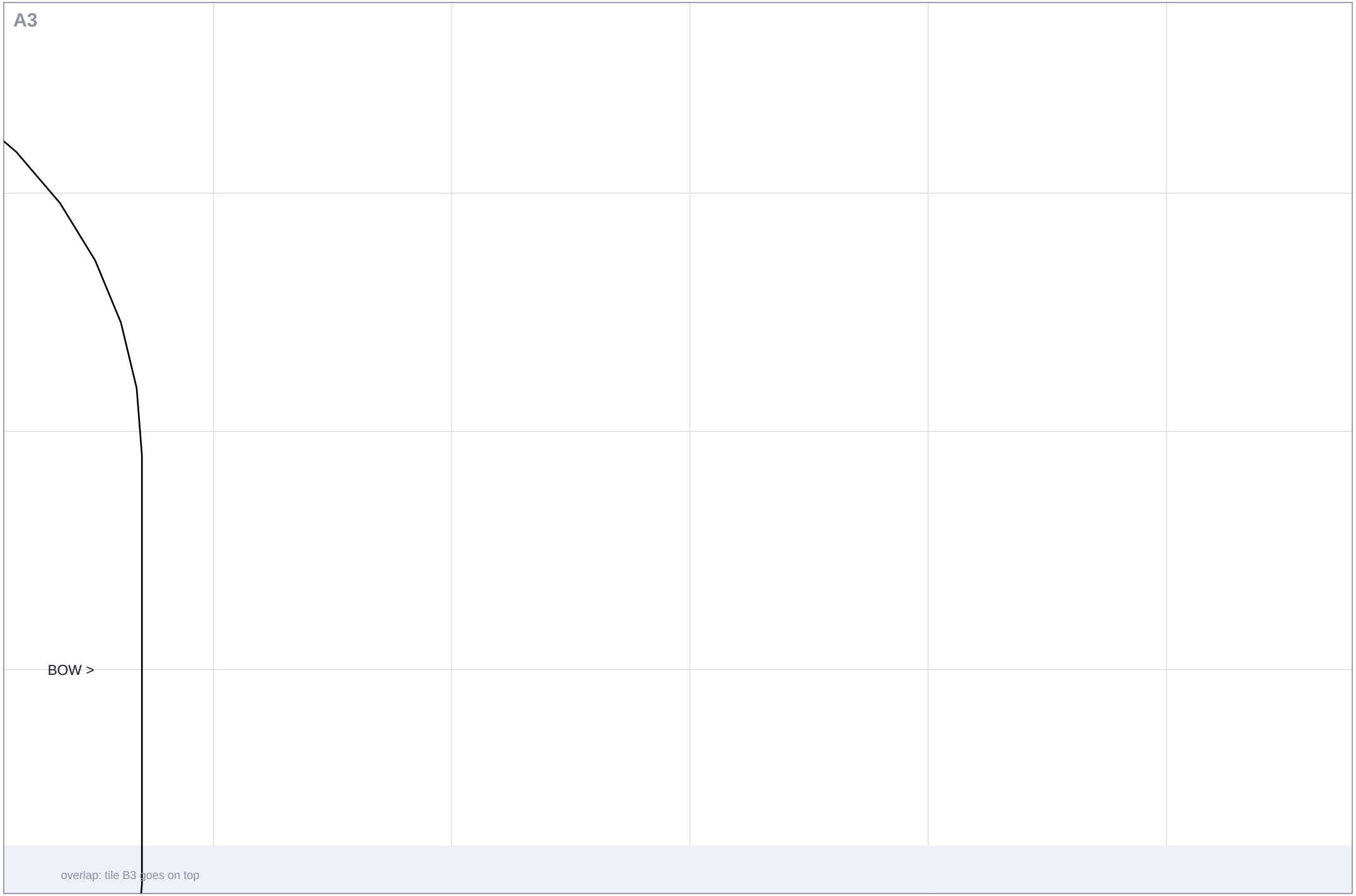
VARUNA AI Pod | Lid top - tile A2 of B3 (3 mm sunboard)

<- must measure 50 mm



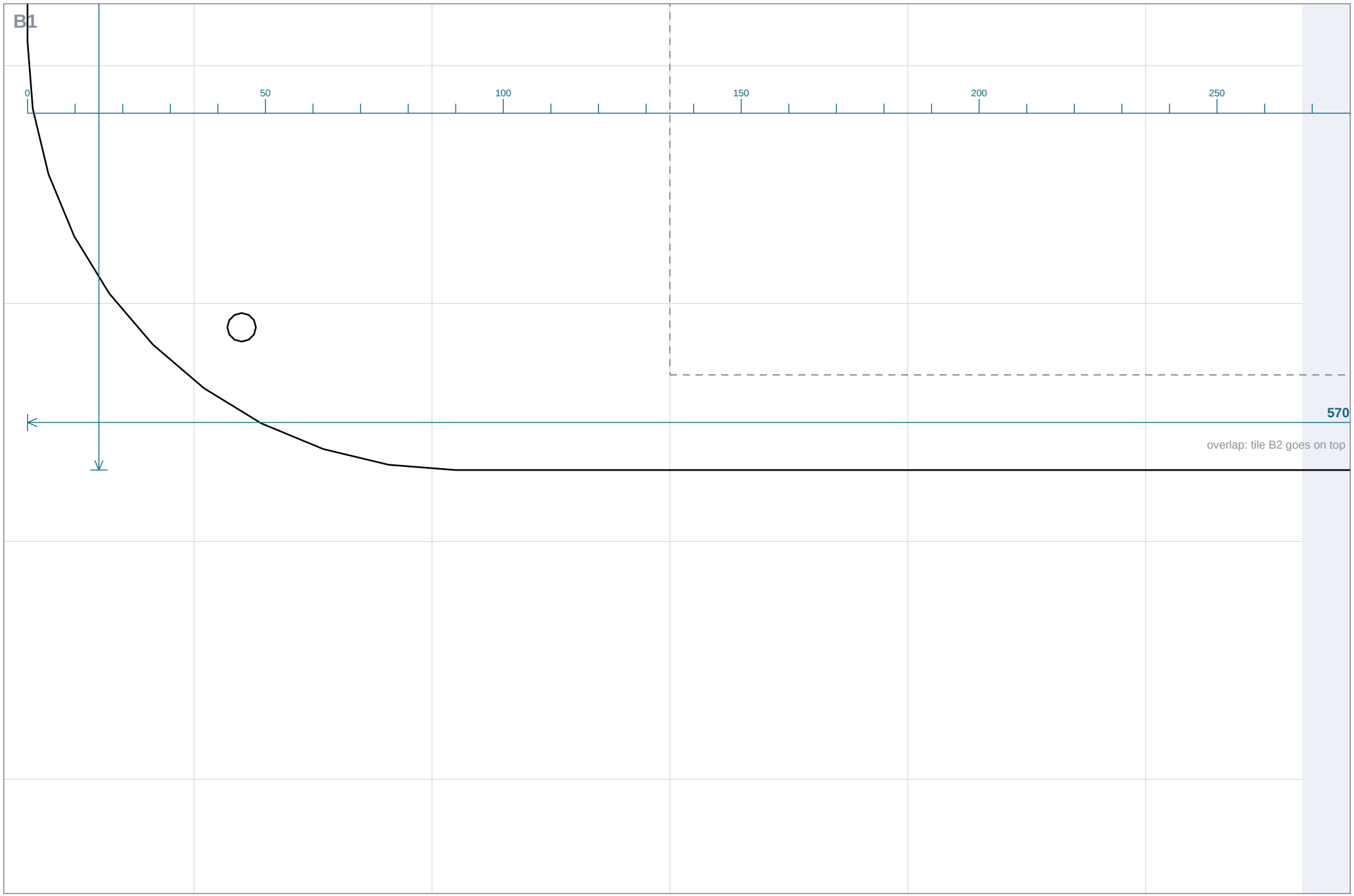
VARUNA AI Pod | Lid top - tile A3 of B3 (3 mm sunboard)

<- must measure 50 mm



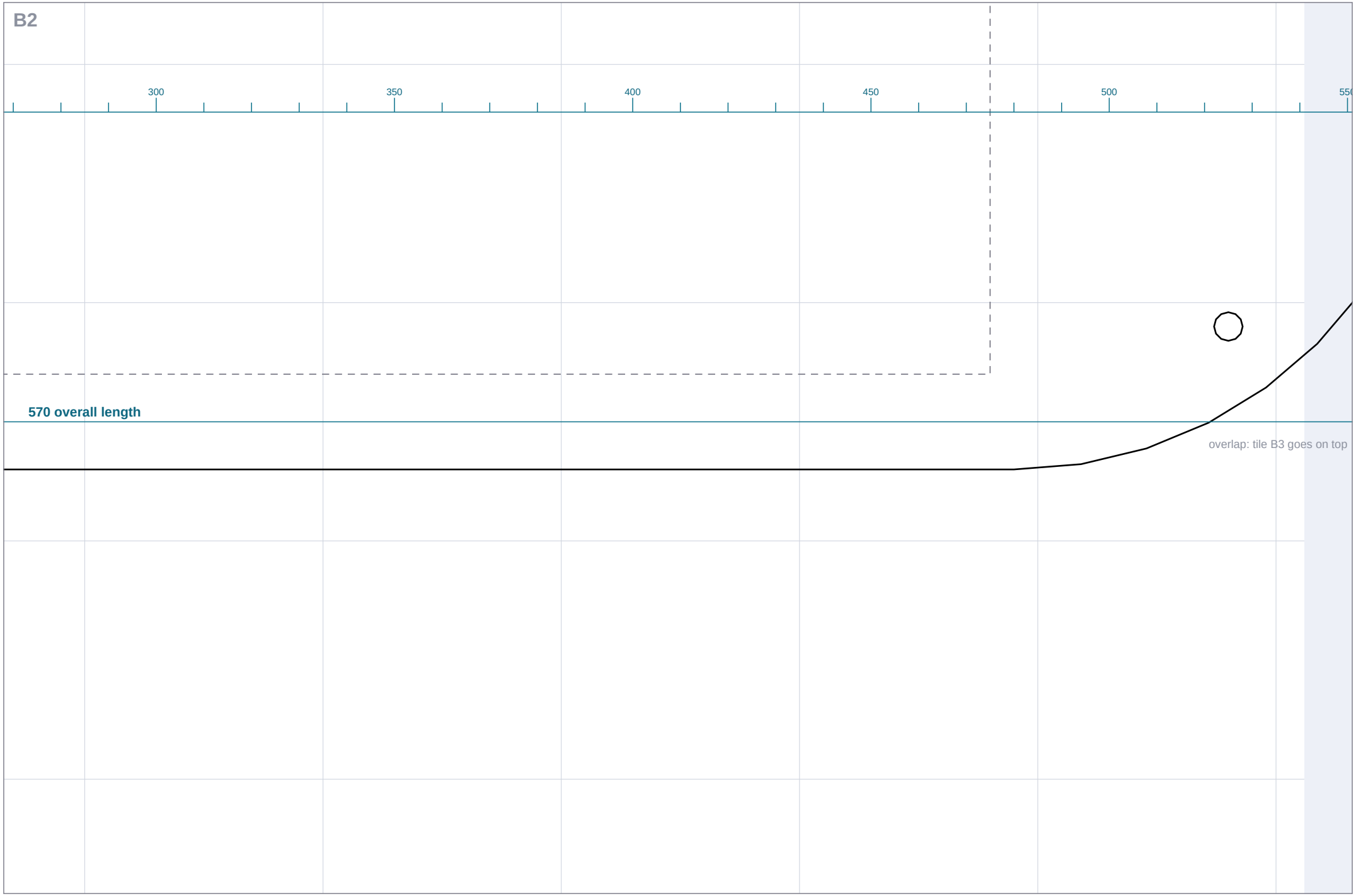
VARUNA AI Pod | Lid top - tile B1 of B3 (3 mm sunboard)

<- must measure 50 mm



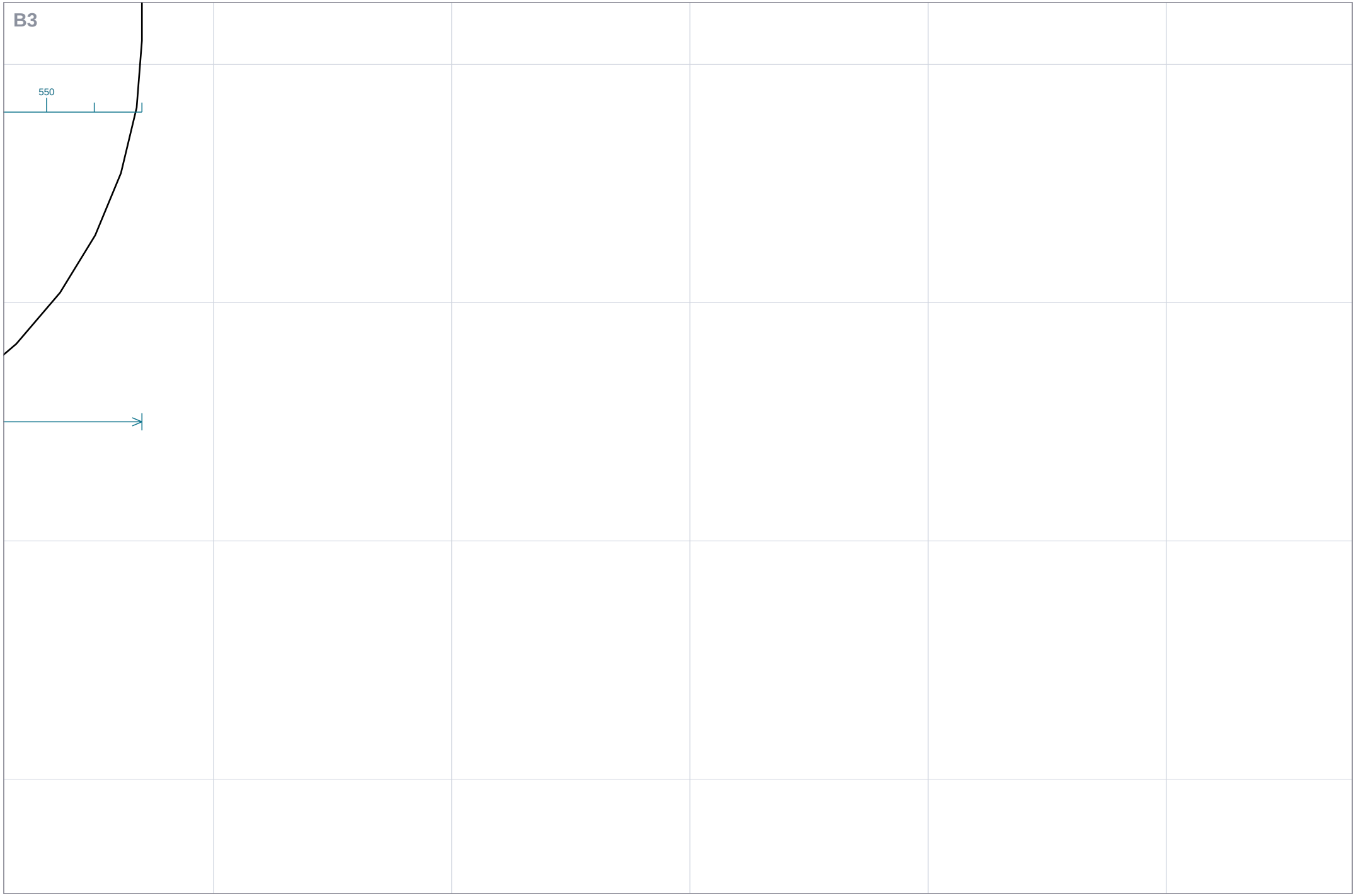
VARUNA AI Pod | Lid top - tile B2 of B3 (3 mm sunboard)

<- must measure 50 mm



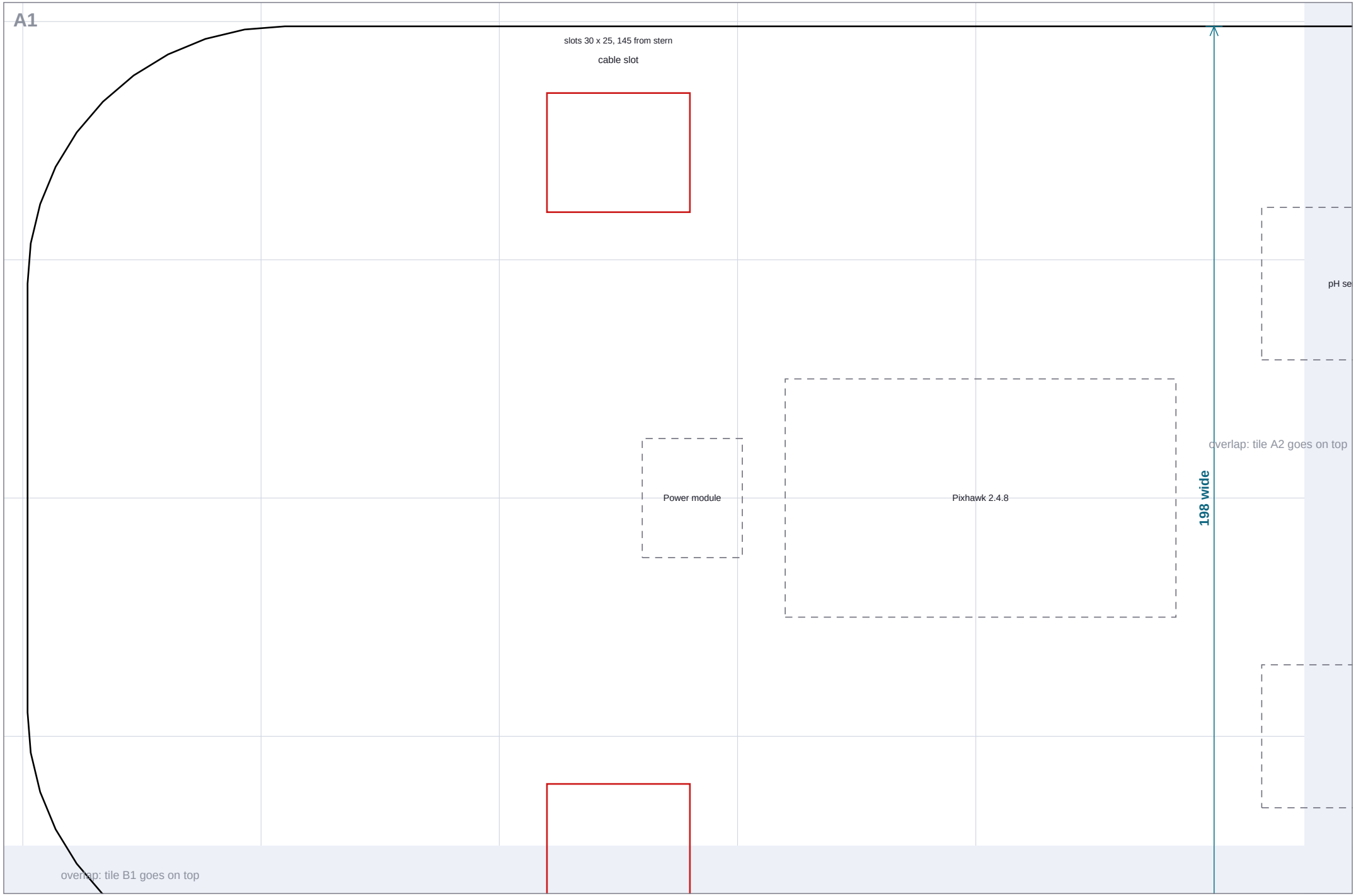
VARUNA AI Pod | Lid top - tile B3 of B3 (3 mm sunboard)

<- must measure 50 mm



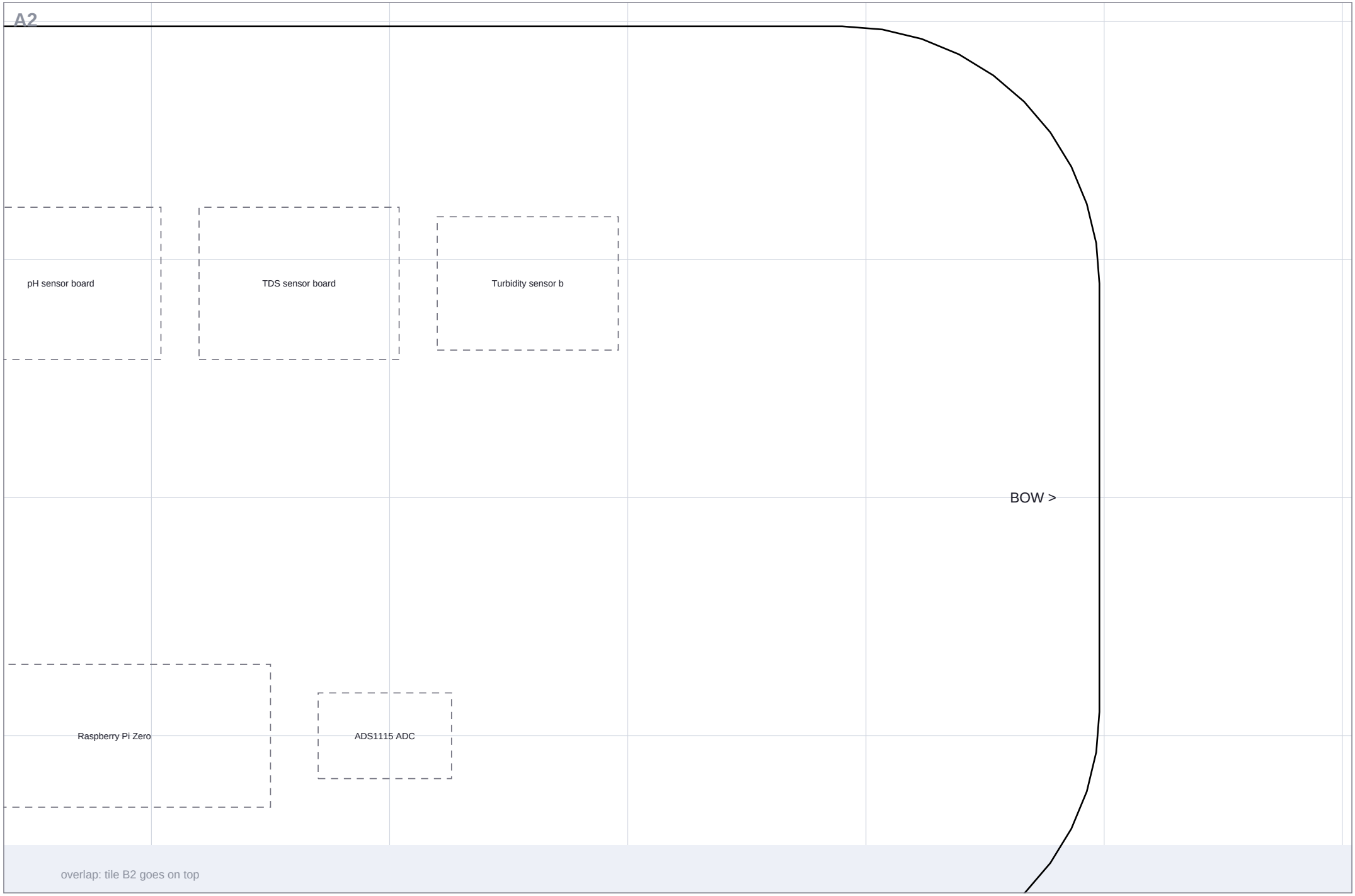
VARUNA AI Pod | Shelf - tile A1 of B2 (3 mm sunboard)

<- must measure 50 mm



VARUNA AI Pod | Shelf - tile A2 of B2 (3 mm sunboard)

<- must measure 50 mm



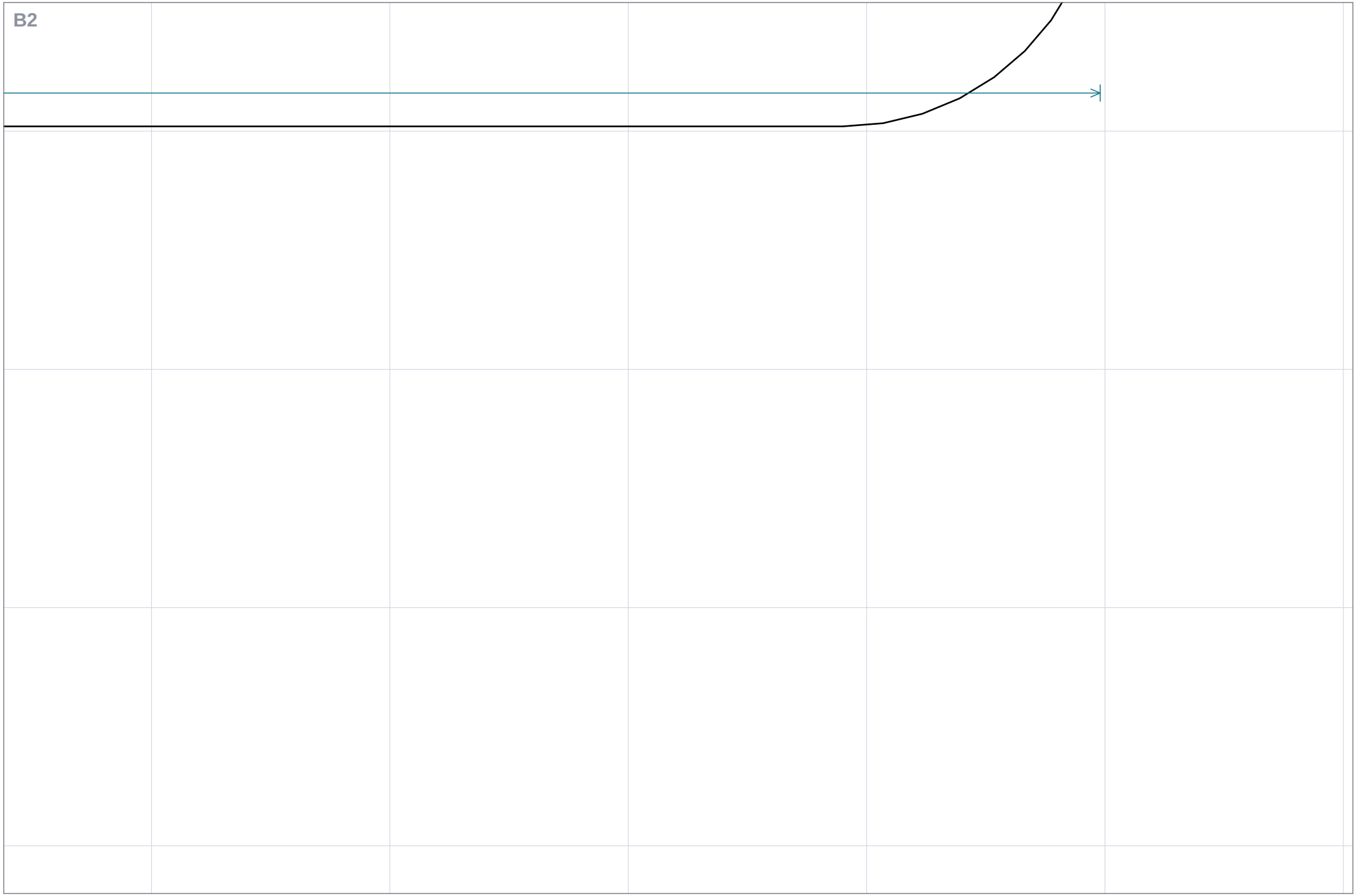
VARUNA AI Pod | Shelf - tile B1 of B2 (3 mm sunboard)

<- must measure 50 mm



VARUNA AI Pod | Shelf - tile B2 of B2 (3 mm sunboard)

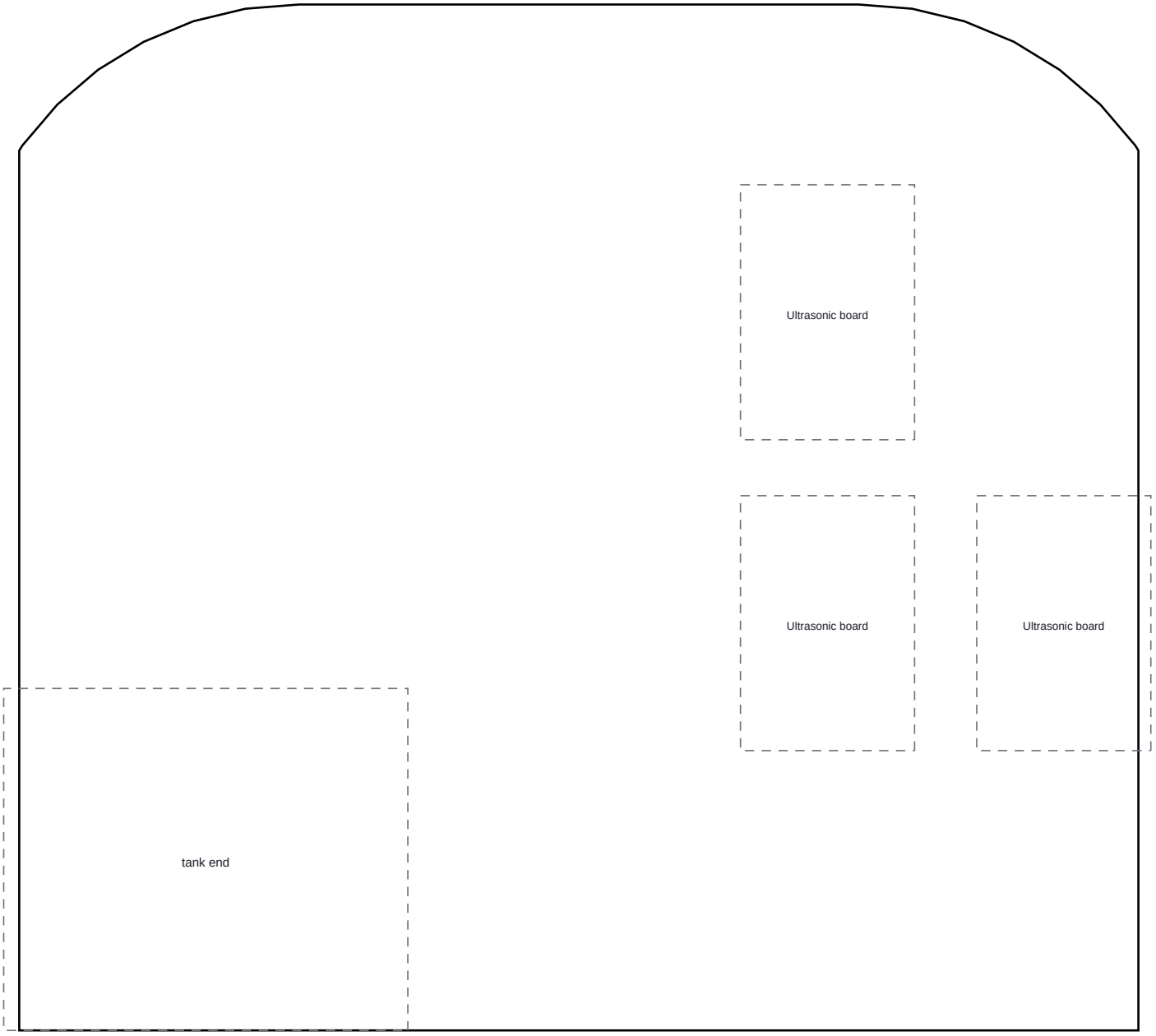
<- must measure 50 mm



<- must measure 50 mm

Drawer tunnel roof

5 mm sunboard | 184 x 165 mm



<- must measure 50 mm

Drawer bottom

3 mm sunboard | 203 x 134 mm

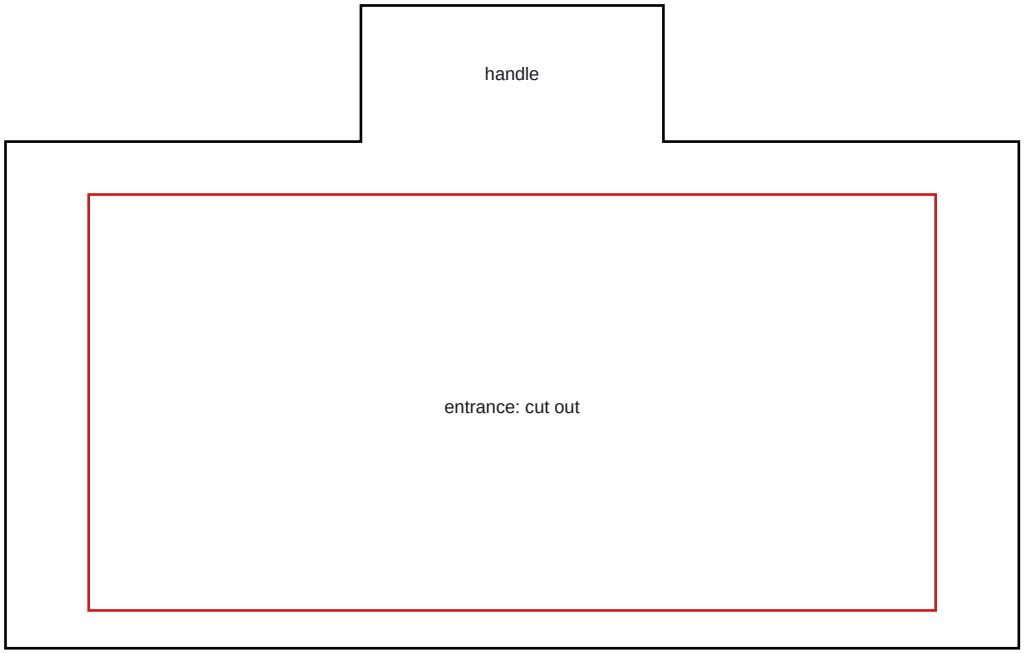


VARUNA AI Pod | small parts 3 of 5

<- must measure 50 mm

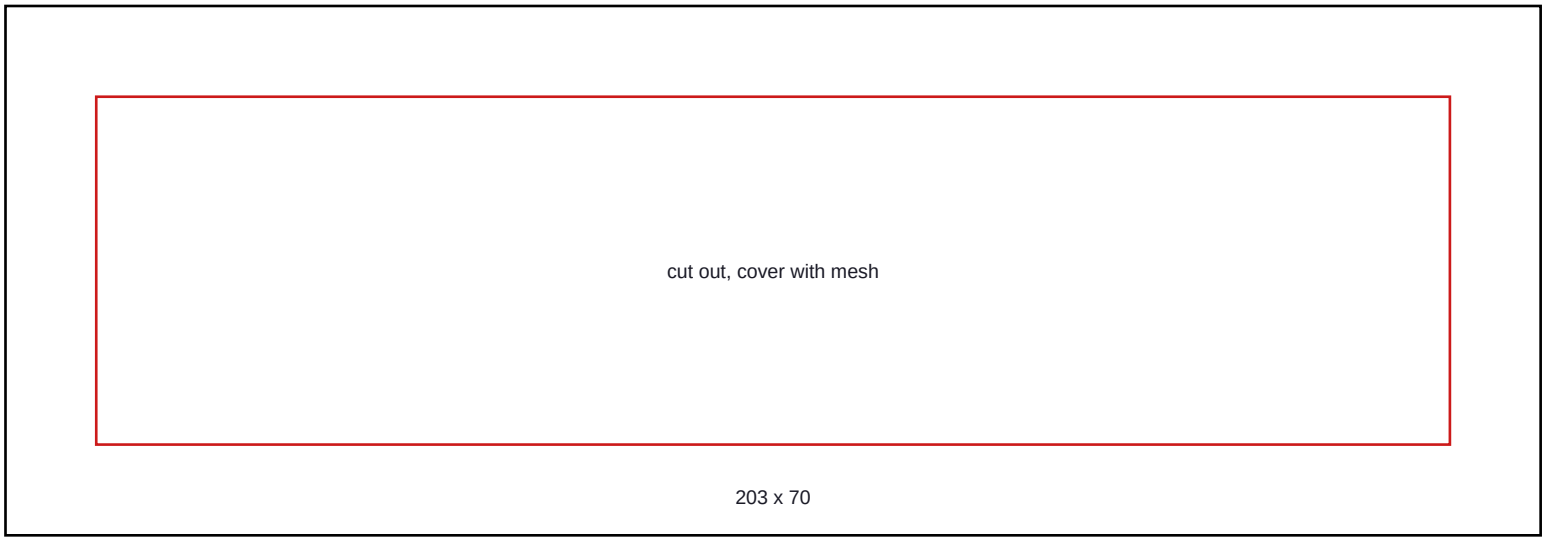
Drawer front (entrance + handle)

3 mm sunboard | 134 x 85 mm



Drawer side (left)

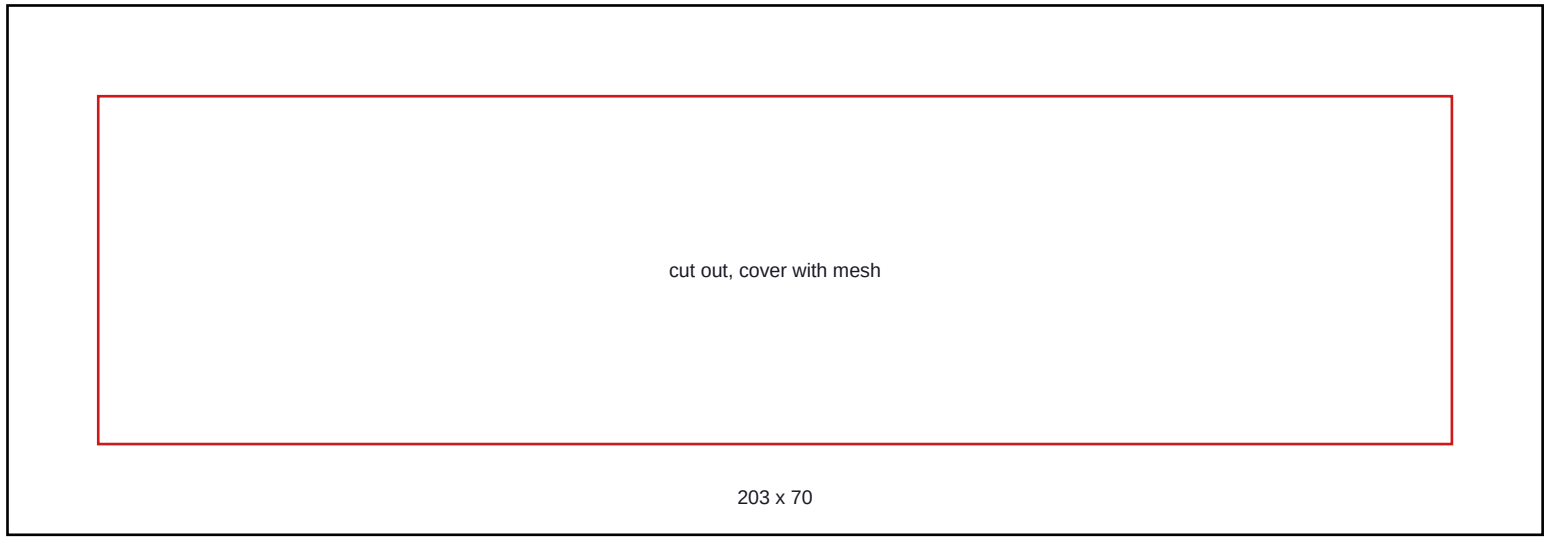
3 mm sunboard | 203 x 70 mm



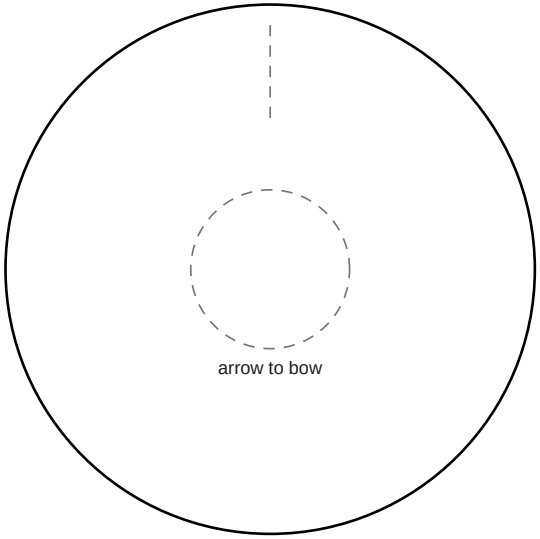
VARUNA AI Pod | small parts 4 of 5

<- must measure 50 mm

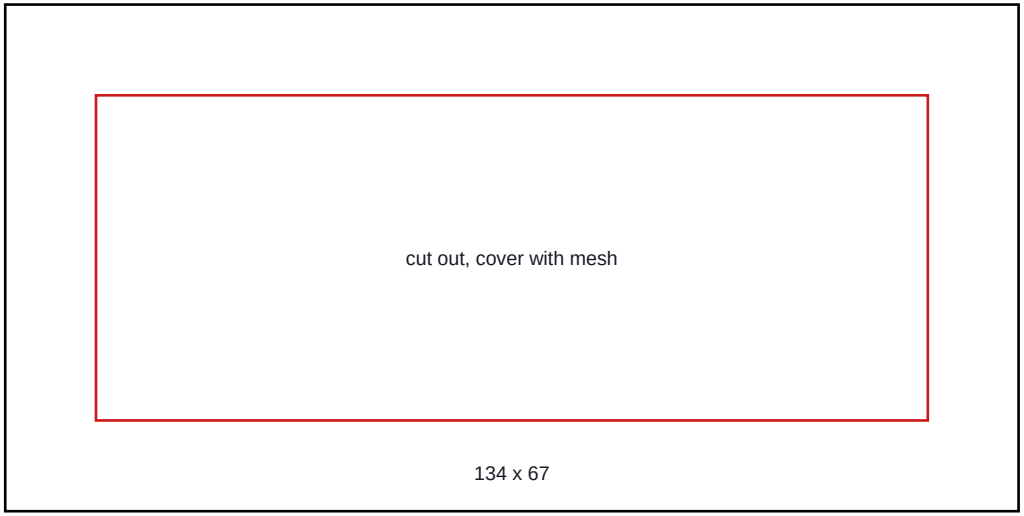
Drawer side (right)
3 mm sunboard | 203 x 70 mm



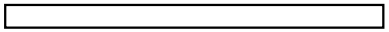
GPS disc
5 mm sunboard | 70 x 70 mm



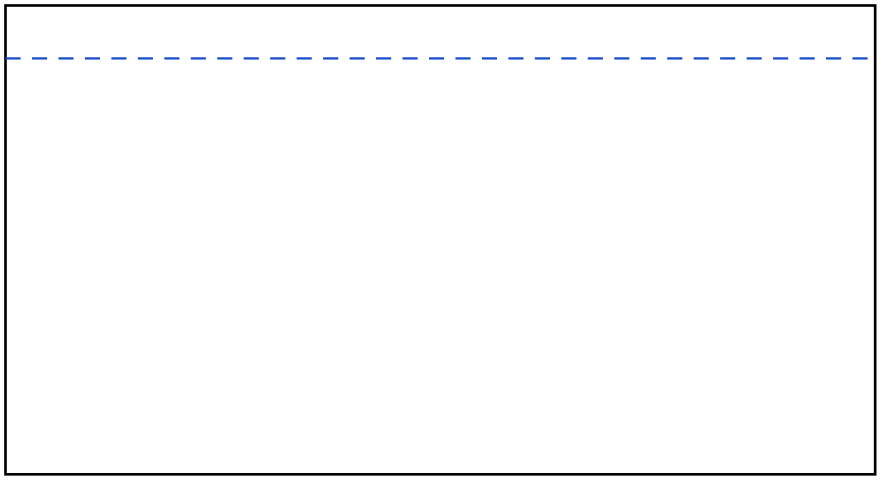
Drawer back
3 mm sunboard | 134 x 67 mm



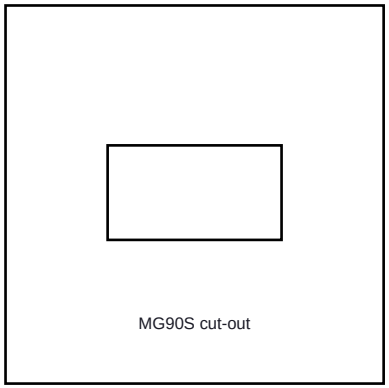
VARUNA AI Pod | small parts 5 of 5



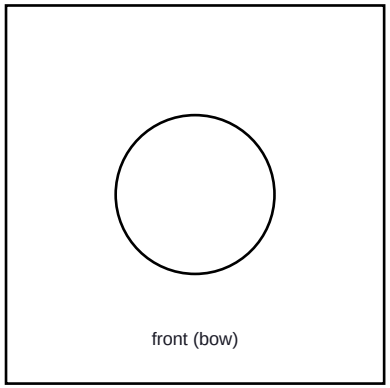
One-way flap
PET sheet | 115 x 62 mm



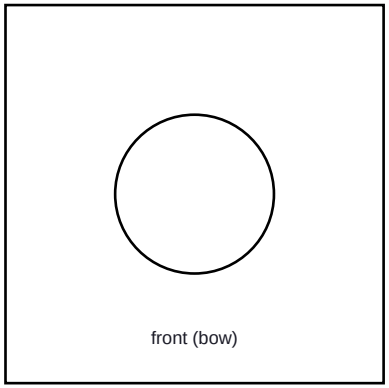
Servo plate
3 mm sunboard | 50 x 50 mm



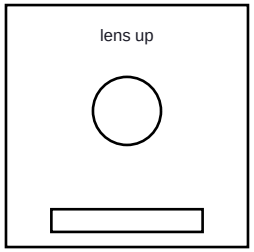
Mast socket layer
5 mm sunboard (cut 2, top one with hole) | 50 x 50 mm



Mast socket layer
5 mm sunboard (cut 2, top one with hole) | 50 x 50 mm



Camera plate
3 mm sunboard | 32 x 32 mm



Camera wedge side (10 deg)
3 mm sunboard | 30 x 6 mm



Camera wedge side (10 deg)
3 mm sunboard | 30 x 6 mm

